

# **Python for Clinical Study Reports and Submission**

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# Welcome

Welcome to Python for Clinical Study Reports and Submission. Clinical study reports (CSRs) sit at the heart of clinical trial development. They bring the design, analysis, and interpretation of a study together in an integrated document.

This book shows how to use Python to plan, build, and automate CSRs, from data and analyses to tables, figures, and listings. We follow the [ICH E3 guidance on structure and content](#) and demonstrate practical, reproducible workflows that help you meet those expectations.

Whether you are new to Python or already working in clinical reporting, you will find small, reusable examples you can adopt right away, along with patterns and checklists for organizing work for regulatory submission and effective team collaboration.

**This is a work-in-progress draft.**

# Preface

## In this book

This book is designed for people who are interested in using Python for clinical development. Each part of the book makes certain assumptions about the readers' background:

- Part 1, titled “Environment and toolchain”, provides general information on setting up Python development environments for clinical reporting.
- Part 2, titled “Clinical trial project”, provides general information and examples on creating tables, listings, and figures using Python. It assumes that readers are individual contributors to a clinical project with prior experience in Python. Familiarity with data manipulation using Polars is expected. Recommended references for this part include [Python Polars: The Definitive Guide](#) and the [rtflite documentation](#).
- Part 3, titled “Analysis package”, provides general information and examples on managing a clinical trial A&R project using Python. It assumes that readers are project leads who have experience in Python package development.
- Part 4, titled “eCTD submission”, provides general information on preparing submission packages related to the CSR in the electronic Common Technical Document (eCTD) format using [pkglite for Python](#). It assumes that readers are project leads of clinical projects who possess experience in Python package development and regulatory submission processes.

## Philosophy

We share the same philosophy described in the introduction of the [R Packages](#) book (Wickham and Bryan 2023), which we quote below:

- “Anything that can be automated, should be automated.”
- “Do as little as possible by hand. Do as much as possible with functions.”

## Authors and contributors

This document is a collaborative effort maintained by a community. As you read through it, you also have the opportunity to contribute and enhance its quality. Your input and involvement play a vital role in shaping the excellence of this document.

- Authors: made significant contributions to at least one chapter, constituting the majority of the content.

[Yilong Zhang](#), [Nan Xiao](#),

## **Part I**

# **Environment and toolchain**

# 1 Python developer setup

## Objective

Set up a productive Python development environment for clinical study reporting. Learn about IDE options, essential extensions, and workflow tools.

## 1.1 Development environments

For this book, you have several options for your development environment. Choose the one that best fits your current setup and constraints.

### 1.1.1 GitHub Codespaces

GitHub Codespaces provides a cloud-based development environment with everything pre-configured. This is the easiest option if you don't have a local Python setup.

We will provide a dev container configuration that includes:

- Python with uv pre-installed.
- Quarto pre-installed for document rendering.
- All necessary VS Code extensions.
- Consistent environment across all readers, useable in the web browser.

To use a pre-configured Codespaces for this book, follow the “Open in GitHub Codespaces” link below.

[https://codespaces.new/nanxstats/pycsr?quickstart=1&devcontainer\\_path=.devcontainer%2Fdevcontainer.json](https://codespaces.new/nanxstats/pycsr?quickstart=1&devcontainer_path=.devcontainer%2Fdevcontainer.json)

Click “Create new Codespace”, you will see a notification “Setting up remote connection: Building codespace...” After two or three minutes, your Codespace will be ready with everything set up.

#### Note

Codespaces currently offers [120 hours of free compute time per month](#) for personal accounts. This is more than sufficient for this book.

#### Caution

Remember to [stop your Codespace](#) after this workshop to avoid unnecessary usage. We will remind you at the end of this book.

### 1.1.2 Positron

Positron is Posit’s next-generation data science IDE, built on Code OSS (the open source core of VS Code), with specific improvements for R and Python development.

Key features for Python work:

- Native notebook support.
- Interactive variable explorer.
- Integrated plot viewer.
- Built-in data viewer for DataFrames.

Download Positron from <https://positron.posit.co/>.

### 1.1.3 VS Code

Visual Studio Code remains the most popular choice for Python development. It offers a rich ecosystem of extensions and tools.

Essential extensions for this book:

- [Python](#): Core Python language support.

Positron uses Open VSX instead of the Microsoft VS Code marketplace. Most essential Python extensions are available, but the selection is more limited.

- [Pylance](#): Fast, feature-rich Python language server.
- [Ruff](#): Lightning-fast Python code linting and formatting.
- [Even Better TOML](#): Syntax highlighter and formatter for TOML files.
- [Quarto](#): Authoring support for Quarto documents.
- [LaTeX Workshop](#): LaTeX authoring support; used here for PDF preview.

## 1.2 VS Code settings

### 1.2.1 Unicode highlighting

Python allows Unicode characters in strings and identifiers. AI coding tools might also generate code with non-ASCII characters. For regulatory work, you should highlight non-ASCII characters to find these hidden issues early and avoid problems in submissions.

#### Via Settings UI:

1. Open Command Palette (Cmd/Ctrl + Shift + P)
2. Search for “Preferences: Open Settings (UI)”
3. Search for “Unicode Highlight”
4. Enable “Non Basic ASCII” for both trusted and untrusted workspaces

#### Via Settings JSON:

Open Command Palette with Cmd/Ctrl + Shift + P, select “Preferences: Open User Settings (JSON)”, then add:

```
"editor.unicodeHighlight.nonBasicASCII": true
```

This highlights characters like curly quotes, em dashes, and other non-ASCII characters that could cause issues in eCTD submission packages.



## 1.3 Terminal setup

For local development, you will interact with uv and Quarto through the terminal.

### 1.3.1 Shell

Any modern shell works well:

- macOS/Linux: zsh (default on macOS), bash
- Windows: PowerShell, Windows Terminal

### 1.3.2 Terminal emulator

If you are on macOS and want a faster terminal experience, consider [Ghostty](#). It is written in Zig for exceptional performance.

## 1.4 AI coding assistants

Modern agentic AI coding tools can accelerate statistical and clinical coding tasks, especially for popular programming languages like Python. We encourage you to use them, for example:

- Codex (command-line interface, VS Code extension)
- Claude Code (command-line interface)
- Cursor (AI-first editor)
- GitHub Copilot (VS Code extension)

### 1.4.1 Effective use of AI tools

To use AI assistants effectively for programming, you need:

**Product manager mindset:** Know exactly what you want to build. In clinical reporting, this means understanding the table shell, statistical method, and regulatory requirements.

**Software architect mindset:** Evaluate model outputs critically. Can you spot issues with data transformations? Do the statistical computations match the statistical analysis plan? Is the output format submission-ready?

 Warning

AI tools are assistants, not replacements for domain expertise. Always verify outputs against statistical analysis plans and regulatory guidance.

## 1.5 What's next

With your development environment configured, you are ready to learn about `uv`, the modern project management tool for Python.

In the next chapter, we will cover:

- Creating and managing Python projects.
- Pinning Python versions.
- Installing dependencies.
- Understanding the modern Python packaging ecosystem.

## 2 Python projects with uv

### Objective

Learn how to use uv to create, manage, and maintain Python projects. Understand virtual environments, dependency management, and the modern Python packaging ecosystem.

### 2.1 Why virtual environments

In Python, virtual environments are not optional. They are essential for any serious project work.

Unlike R's renv (which primarily helps with reproducibility), Python virtual environments serve a fundamental purpose: **isolating project dependencies from the system Python**.

Here is why this matters:

- Different projects need different package versions.
- System Python library should never be modified directly.
- Dependency conflicts are common and destructive.
- Reproducibility requires exact version control.

### Warning

Installing packages globally with `pip install` without a virtual environment will cause conflicts and break system tools. Always use virtual environments. To install Python packages as global command-line tools, use `pipx`.

## 2.2 What is uv

uv is a modern Python package and project manager written in Rust. It replaces and improves upon a scattered toolchain:

- `pip` (package installation)
- `venv` (virtual environment creation)
- `pyenv` (Python version management)
- `pip-tools` (dependency locking)
- `setuptools` (package building)

Benefits of uv:

- **Fast:** 10-100x faster than pip due to Rust implementation.
- **Complete:** Manages Python versions, dependencies, and builds.
- **Modern:** Uses `pyproject.toml` as the single source of truth.
- **Reliable:** Automatic dependency resolution and lock files.

In R terms, uv combines functionality from `renv`, `devtools`, `usethis`, and `pak` into a single, cohesive tool.

## 2.3 Python packaging standards

Python has standardized on `pyproject.toml` as the configuration file for all projects. This is similar to R's `DESCRIPTION` file but uses TOML format.

The official Python packaging guide is available at <https://packaging.python.org/>.

Key concepts:

- `pyproject.toml` defines project metadata and dependencies.
- `uv.lock` records exact versions (like `renv.lock`).
- Build backends (like `uv_build`) create distributable packages.

## 2.4 Installing uv

Follow the [official installation guide](#).

**macOS and Linux:**

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

**Windows:**

```
powershell -ExecutionPolicy Bypass -c "irm https://astral.sh/uv/install.ps1 | iex"
```

**Via Homebrew (macOS):**

```
brew install uv
```

Verify installation:

```
uv --version
```

## 2.5 Updating uv

uv can update itself:

```
uv self update
```

Regular updates are important because uv frequently adds support for new Python versions and features.

### **i** Note

uv uses Python distributions from the [python-build-standalone](#) project. These are optimized, portable Python builds that work consistently across platforms.

## 2.6 Initialize a project

⚠ For GitHub Codespaces users

If you are using GitHub Codespaces, the `pycsr` project folder is opened by default. Before creating a new practice project, close this folder via **File > Close Folder** so your shell returns to the home directory. This prevents `uv` from getting confused about which `uv.lock` file to write when you “initialize a new project within an existing project”.

Create a new Python project:

```
uv init pycsr-example
cd pycsr-example
```

By default, `uv init` creates a packaged project with a `src/` layout and configures `uv_build` as its build backend. For a project that only needs an environment and standalone scripts, use `uv init --no-package`.

This creates a basic structure:

```
pycsr-example/
├── .python-version      # Pinned Python version
├── pyproject.toml       # Project metadata and dependencies
├── README.md           # Project documentation
└── src/
    ├── pycsr_example/
    └── __init__.py
```

### 2.6.1 Project structure

The `pyproject.toml` file contains project configuration:

```
[project]
name = "pycsr-example"
version = "0.1.0"
description = "Example clinical study report project"
dependencies = []
```

Notice the directory name uses hyphens (`pycsr-example`) while the package name uses underscores (`pycsr_example`). This is Python convention.

```
[build-system]
requires = ["uv_build>=0.12.5,<0.13.0"]
build-backend = "uv_build"
```

The generated `uv_build` version range requirement is based on the installed `uv` version, so keep the range written by `uv init`.

Key sections:

- `[project]`: Package metadata.
- `[project.dependencies]`: Hard, runtime dependencies.
- `[dependency-groups.dev]`: Development dependencies.
- `[build-system]`: How to build the package.

## 2.7 Pin Python version

Specify the exact Python version for your project:

```
uv python pin 3.14.0
```

This updates `.python-version` file so everyone uses the same Python version when they restore the environment.

### ! Important

Use the full `MAJOR.MINOR.PATCH` version (for example, `3.14.0`) rather than just `MAJOR.MINOR` (for example, `3.14`). This prevents drift as new patch versions are released.

Why pin the exact version:

- Patch releases can introduce subtle behavior changes.
- Reproducibility requires exact version matching.
- Regulatory submissions should document the exact Python version.

Check which Python versions are available:

```
uv python list
```

Install a specific Python version if needed:

```
uv python install 3.14.0
```

## 2.8 Managing dependencies

### 2.8.1 Adding dependencies

Add runtime dependencies:

```
uv add polars plotnine rtflite
```

Add development-only dependencies:

```
uv add --dev ruff pytest mypy
```

This updates `pyproject.toml`:

```
[project]
dependencies = [
    "plotnine>=0.15.1",
    "polars>=1.35.2",
    "rtflite>=1.1.0",
]

[dependency-groups]
dev = [
    "mypy>=1.18.2",
    "pytest>=9.0.1",
    "ruff>=0.14.5",
]
```



### **i** Note

By default, uv adds dependencies with `>=` constraints. This allows updates within compatible versions. The lock file ensures exact versions are used.

## **2.8.2 Removing dependencies**

Remove a package:

```
uv remove pandas
```

This removes the package from both `pyproject.toml` and the environment.

## **2.9 Lock files and syncing**

### **2.9.1 Creating and updating the lock file**

Generate or update the lock file:

```
uv sync
```

This creates `uv.lock`, which records:

- Exact version of every package.
- All transitive dependencies.
- Package hashes for verification.

The lock file ensures reproducibility across different machines and over time.

### **2.9.2 Upgrading dependencies**

To update packages while respecting constraints in `pyproject.toml`:

```
uv lock --upgrade
```

Then synchronize the environment:

```
uv sync
```

This is similar to:

- R: `renv::update()` followed by `renv::snapshot()`.
- Node.js: `npm update` followed by `npm install`.

The two-step process (lock & sync) gives you control: you can review lock file changes before updating your environment.

## 2.10 Running commands

You have two options for running commands in your project environment.

### 2.10.1 Option 1: Activate the virtual environment

```
source .venv/bin/activate # macOS/Linux  
# or  
.venv\Scripts\activate    # Windows
```

Then run commands directly:

```
python -m pycsr_example  
pytest  
ruff check
```

Deactivate when done:

```
deactivate
```

### 2.10.2 Option 2: Use `uv run`

Run commands without activation:

```
uv run python -m pycsr_example
uv run pytest
uv run ruff check
```

#### Tip

`uv run` is convenient for one-off commands and CI/CD scripts. For interactive work, activating the environment is often more ergonomic.

### 2.10.3 `uv run` and `uvx`

`uvx` runs tools in isolated, temporary environments:

```
uvx ruff check .
uvx black --check .
```

Use `uvx` when:

- Running tools you don't want to install in the project.
- Trying packages without adding them as dependencies.
- Running scripts that declare their own dependencies.

Use `uv run` when:

- Running project code.
- Running tests.
- Using project dependencies.

See [using tools in uv](#) for details.

## 2.11 Building and publishing

Packaged projects created with `uv init` are ready to build with `uv build`:

```
[build-system]
requires = ["uv_build>=0.12.5,<0.13.0"]
build-backend = "uv_build"
```

### 2.11.1 Build wheel

Create distribution files:

```
uv build
```

This creates:

- `dist/pycsr_example-0.1.0.tar.gz` (source distribution)
- `dist/pycsr_example-0.1.0-py3-none-any.whl` (wheel)

### 2.11.2 Publish to PyPI

Publish to the Python Package Index:

```
uv publish
```

#### Note

Building and publishing are not typically needed for internal clinical reporting projects. However, if you develop reusable tools like table generation packages, open sourcing in a GitHub repository and publishing on PyPI will make them more visible.

## 2.12 Exercise

Create a small project to practice uv commands:

1. Initialize a new project called `csr-practice`.
2. Pin Python to version 3.14.0 (or latest available).
3. Add `polars` as a dependency.
4. Add `pytest` as a development dependency.
5. Examine the generated `pyproject.toml` and `uv.lock` files.
6. Run Python using `uv run python --version`.

View solution

```
# Initialize project
uv init csr-practice
cd csr-practice

# Pin Python version
uv python pin 3.14.0

# Add dependencies
uv add polars
uv add --dev pytest

# View configuration
cat pyproject.toml

# Check lock file
cat uv.lock

# Run Python
uv run python --version
```

Your `pyproject.toml` should look similar to:

```
[project]
name = "csr-practice"
version = "0.1.0"
description = "Add your description here"
```

```
dependencies = [  
    "polars>=1.18.0",  
]  
  
[dependency-groups]  
dev = [  
    "pytest>=8.3.4",  
]  
  
[build-system]  
requires = ["uv_build>=0.12.5,<0.13.0"]  
build-backend = "uv_build"
```

## 2.13 What's next

Now that you understand uv basics, the next chapter covers the Python package toolchain:

- Formatting and linting with Ruff.
- Type checking with mypy.
- Testing with pytest.
- Documentation generation.
- Development workflows for clinical reporting.

## 3 Python package toolchain

### Objective

Learn the essential development tools for Python projects: formatting, linting, type checking, testing, and documentation. Build a professional development workflow for clinical reporting.

### 3.1 The modern Python toolchain

In R, packages like `devtools`, `usethis`, `styler`, `lintr`, and `testthat` provide development infrastructure. Python's ecosystem distributes these functions across specialized tools.

For clinical reporting projects, we recommend:

- **uv**: Package and environment management.
- **Ruff**: Code formatting and linting.
- **mypy**: Static type checking.
- **pytest**: Unit testing framework.
- **quartodoc**: Documentation and reporting.

All tools are installed as development dependencies and configured through `pyproject.toml`.

For R users, think of this as: `uv = renv + pak + devtools`, `Ruff = styler + lintr`, `pytest = testthat`, `mypy =` (no direct R equivalent).

### 3.2 Ruff: Formatting and linting

Ruff is a super fast linter and formatter written in Rust. It replaces multiple legacy tools (Black, isort, Flake8, pyupgrade) with a single, consistent interface.

### 3.2.1 Installation

Add Ruff as a development dependency:

```
uv add --dev ruff
```

### 3.2.2 Code formatting

Format your code:

```
uv run ruff format
```

Or using uvx:

```
uvx ruff format
```

Ruff format:

- Enforces consistent style (like Black).
- Sorts imports automatically.
- Removes trailing whitespace.
- Ensures consistent line lengths.

### 3.2.3 Linting

Check for linting issues:

```
uv run ruff check
```

Fix auto-fixable issues:

```
uv run ruff check --fix
```

Ruff detects:

- Unused imports and variables.
- Undefined names.
- Style violations.
- Common anti-patterns.
- Security issues.



### 3.2.4 Configuration

Add Ruff configuration to `pyproject.toml`:

```
[tool.ruff]
line-length = 88
target-version = "py313"

[tool.ruff.format]
quote-style = "double"
indent-style = "space"

[tool.ruff.lint]
select = [
    "E",      # pycodestyle
    "F",      # Pyflakes
    "UP",     # pyupgrade
    "B",      # flake8-bugbear
    "SIM",    # flake8-simplify
    "I",      # isort
]
ignore = []
```

#### Note

Line length of 88 characters is the Python community standard. It balances readability with modern screen sizes.

## 3.3 Type checking with mypy

Python supports optional type annotations through [PEP 484](#). Type annotations improve code clarity and catch errors before runtime.

### 3.3.1 Why type checking matters

For clinical programming:

- Catch data transformation errors at development time.
- Document expected DataFrame structures.
- Improve IDE autocomplete and refactoring.
- Reduce runtime errors in production.

### 3.3.2 Installation

Add mypy as a development dependency:

```
uv add --dev mypy
```

### 3.3.3 Basic usage

Check types in your code:

```
uv run mypy .
```

### 3.3.4 Type annotation example

Without types:

```
def calculate_bmi(weight, height):  
    return weight / (height ** 2)
```

With types:

```
def calculate_bmi(weight: float, height: float) -> float:  
    """Calculate BMI from weight (kg) and height (m)."""  
    return weight / (height ** 2)
```

The type checker verifies:

- Arguments are the correct type.
- Return value matches the declared type.
- Operations are valid for the types used.

### 3.3.5 Configuration

Add mypy settings to `pyproject.toml`:

```
[tool.mypy]
python_version = "3.14"
warn_return_any = true
warn_unused_configs = true
disallow_untyped_defs = false
disallow_incomplete_defs = true
check_untyped_defs = true
no_implicit_optional = true
```

Start with lenient settings  
(`disallow_untyped_defs = false`)  
and progressively tighten as you add  
type annotations to your codebase.

### 3.3.6 Type stubs for libraries

Some libraries don't include type information. Install type stubs when available:

```
uv add --dev types-tabulate
```

#### Note

Popular data science libraries like `polars` include built-in type annotations. Older libraries like `pandas` require separate stub packages (`pandas-stubs`).

## 3.4 Testing with pytest

`pytest` is Python's de facto standard testing framework. It's more powerful and ergonomic than the built-in `unittest` module.

### 3.4.1 Installation

Add pytest and coverage tools:

```
uv add --dev pytest pytest-cov
```

### 3.4.2 Writing tests

Create a `tests/` directory:

```
pycsr-example/  
  src/  
    pycsr_example/  
      __init__.py  
  tests/  
    test_calculations.py
```

Write a simple test in `tests/test_calculations.py`:

```
from pycsr_example.calculations import calculate_bmi  
import pytest  
  
def test_calculate_bmi():  
    # Normal BMI calculation  
    assert calculate_bmi(70, 1.75) == pytest.approx(22.857142857142858)  
  
def test_calculate_bmi_underweight():  
    # BMI < 18.5 indicates underweight  
    assert calculate_bmi(50, 1.75) < 18.5
```

### 3.4.3 Running tests

Run all tests:

```
uv run pytest
```

Run with verbose output:

```
uv run pytest -v
```

Run specific test file:

```
uv run pytest tests/test_calculations.py
```

### 3.4.4 Code coverage

Generate coverage report:

```
uv run pytest --cov=pycsr_example --cov-report=term
```

Generate HTML coverage report:

```
uv run pytest --cov=pycsr_example --cov-report=html
```

This creates `htmlcov/index.html` showing which lines are tested.

#### ! Important

For regulatory submissions, high test coverage demonstrates code quality. Aim for >80% coverage for critical data transformation and statistical computation functions.

### 3.4.5 pytest configuration

Add pytest settings to `pyproject.toml`:

```
[tool.pytest.ini_options]
testpaths = ["tests"]
python_files = ["test_*.py"]
python_functions = ["test_*"]
addopts = [
    "--strict-markers",
    "--strict-config",
    "-ra",
]
```

## 3.5 Documentation generation

For clinical reporting projects, documentation serves two purposes:

1. **Code documentation:** Function and module documentation.
2. **Report generation:** Analysis reports and TLFs.

### 3.5.1 Quarto for reports

We use Quarto for creating reproducible analysis documents:

```
# Install Quarto separately (not via uv)
# See: https://quarto.org/docs/get-started/
```

Quarto documents (`.qmd` files) combine:

- Markdown text.
- Python code cells.
- Generated outputs (tables, listings, figures).

This book itself is written in Quarto.

### 3.5.2 quartodoc for API documentation

For packages that need API documentation (similar to R's `pkgdown`), use `quartodoc`:

```
uv add --dev quartodoc
```

`quartodoc` generates documentation from docstrings and integrates with Quarto for full website generation.

For analysis projects (rather than reusable packages), Quarto alone is usually sufficient. Use `quartodoc` when building analysis packages for team to collaborate on.

## 3.6 Development workflow

Putting it all together, a typical development cycle looks like:

1. **Format code:** `uv run ruff format`
2. **Check linting:** `uv run ruff check --fix`
3. **Verify types:** `uv run mypy .`
4. **Run tests:** `uv run pytest --cov=pycsr_example`
5. **Generate reports:** `quarto render`

### 3.6.1 Pre-commit automation

You can automate these checks using Git hooks (not covered in this book), but manual execution provides better learning and control during development.

## 3.7 Clinical project structure guidelines

In case you need clinical reporting projects using both R and Python:

**Separate R and Python directories:**

```
project/
  r-package/          # R package for R-based analyses
    DESCRIPTION
    R/
    tests/
  python-package/     # Python package for Python-
based analyses
  pyproject.toml
  src/
  tests/
  data/              # Shared input data (SDTM, ADaM)
  output/            # Shared output (TLFs, reports)
```

**Why separate?** There are a few reasons:

- Different build systems.

- Different dependency management.
- Different testing frameworks.
- Different IDE configurations.

#### Shared resources:

- Input datasets (SDTM, ADaM) can be in a common **data/** directory.
- Output deliverables can go to a common **output/** directory.
- Documentation can reference both implementations.

#### Note

For this book, we focus exclusively on Python. Mixed R/Python workflows are beyond scope but follow the same principles.

## 3.8 Exercise

Set up a complete development environment:

1. Create a new project with `uv init dev-practice`.
2. Add development dependencies: `ruff`, `mypy`, `pytest`, `pytest-cov`.
3. Create a simple function in `src/dev_practice/stats.py`:

```
def mean(values: list[float]) -> float:
    return sum(values) / len(values)
```

4. Write a test in `tests/test_stats.py`.
5. Run Ruff format and check.
6. Run mypy type checking.
7. Run pytest with coverage.

[View solution](#)



```

# Create project
uv init dev-practice
cd dev-practice

# Add dev dependencies
uv add --dev ruff mypy pytest pytest-cov

# Create stats module
mkdir -p src/dev_practice
touch src/dev_practice/__init__.py
cat > src/dev_practice/stats.py << 'EOF'
def mean(values: list[float]) -> float:
    """Calculate the arithmetic mean of a list of numbers."""
    if not values:
        raise ValueError("Cannot calculate mean of empty list")
    return sum(values) / len(values)
EOF

# Create test file
mkdir -p tests
cat > tests/test_stats.py << 'EOF'
import pytest
from dev_practice.stats import mean

def test_mean_basic():
    assert mean([1.0, 2.0, 3.0]) == 2.0

def test_mean_single_value():
    assert mean([5.0]) == 5.0

def test_mean_empty_raises():
    with pytest.raises(ValueError):
        mean([])
EOF

# Run checks
uv run ruff format .
uv run ruff check .
uv run mypy src/
uv run pytest --cov=dev_practice --cov-report=term

```

Expected output from pytest:

```
===== test session starts =====
platform darwin -- Python 3.14.0, pytest-9.0.1, pluggy-
1.6.0
rootdir: /Users/user/dev-practice
configfile: pyproject.toml
plugins: cov-7.0.0
collected 3 items

tests/test_stats.py ... [100%]

===== tests coverage =====
----- coverage: platform darwin, python 3.14.0-
final-0 -----

Name                                Stmts  Miss  Cover
-----
src/dev_practice/__init__.py         0      0  100%
src/dev_practice/stats.py            4      0  100%
-----
TOTAL                                4      0  100%
===== 3 passed in 0.02s =====
```

### 3.9 Example repositories

Demo project repositories have been created:

- **Python package example:** <https://github.com/elong0527/demo-py-esub>
- **eCTD package example:** <https://github.com/elong0527/demo-py-ectd>

With the knowledge from this chapter, you can understand how these projects are organized and develop similar professional Python packages for clinical reporting.

## 3.10 What's next

You now have a complete Python development environment with:

- `uv` for project and dependency management.
- `Ruff` for code quality.
- `mypy` for type safety.
- `pytest` for testing.
- `Quarto` for documentation.

Next part will introduce how to create real clinical study reports, demonstrating TLF generation with `polars` and `rtflite`.

## **Part II**

# **Clinical trial project**

## 4 TLF overview

### Objective

Understand the regulatory context and importance of TLFs in clinical study reports. Learn basic concepts of Polars and rtfite for TLF generation.

### 4.1 Overview

Tables, listings, and figures (TLFs) are essential components of clinical study reports (CSRs) and regulatory submissions. Following [ICH E3 guidance](#), TLFs provide standardized summaries of clinical trial data that support regulatory decision-making.

This chapter provides an overview of creating TLFs using Python, focusing on the tools and workflows demonstrated throughout this book.

### 4.2 Background

Submitting clinical trial results to regulatory agencies is a crucial aspect of clinical development. The [Electronic Common Technical Document \(eCTD\)](#) has emerged as the global standard format for regulatory submissions. For instance, the United States Food and Drug Administration (US FDA) [mandates the use of eCTD](#) for new drug applications and biologics license applications.

A CSR provides comprehensive information about the methods and results of an individual clinical study. To support the statistical analysis, numerous tables, listings, and figures are included within the main text and appendices. The creation of a

CSR is a collaborative effort that involves various professionals such as clinicians, medical writers, statisticians, and statistical programmers.

Within an organization, these professionals typically collaborate to define, develop, validate, and deliver the necessary TLFs for a CSR. These TLFs serve to summarize the efficacy and/or safety of the pharmaceutical product under study. In the pharmaceutical industry, Microsoft Word is widely utilized for CSR preparation. As a result, the deliverables from statisticians and statistical programmers are commonly provided in formats such as `.rtf`, `.doc`, `.docx` to align with industry standards and requirements.

#### Note

Each organization may define specific TLF format requirements that differ from the examples in this book. It is advisable to consult and adhere to the guidelines and specifications set by your respective organization when preparing TLFs for submission.

By following the ICH E3 guidance, most TLFs in a CSR are located at:

- Section 10: Study participants
- Section 11: Efficacy evaluation
- Section 12: Safety evaluation
- Section 14: Tables, listings, and figures referenced but not included in the text
- Section 16: Appendices

## 4.3 Datasets

The dataset structure follows [CDISC Analysis Data Model \(ADaM\)](#).

In this project, we use publicly available CDISC pilot study data, which is accessible through the [CDISC GitHub repository](#).

We have converted these datasets from the `.xpt` format to the `.parquet` format for ease of use and compatibility with Python tools. The dataset structure adheres to the CDISC [Analysis Data Model \(ADaM\)](#) standard.

## 4.4 Tools

To exemplify the generation of TLFs in RTF format, we rely on the functionality provided by two Python packages:

- **Polars**: Preparation of datasets in a format suitable for reporting purposes. Polars offers a comprehensive suite of tools and functions for data manipulation and transformation, ensuring that the data is structured appropriately.
- **rtflite**: Creation of RTF files. The `rtflite` package offers functions specifically designed for generating RTF files, allowing us to produce TLFs in the desired format.

## 4.5 Polars

Polars is an open-source library for data manipulation implemented in Rust with Python bindings. It offers exceptional performance while maintaining a user-friendly interface for interactive data analysis.

Key advantages of Polars include:

- **Performance**: 10-100x faster than pandas for most operations due to Rust implementation
- **Memory efficiency**: Lazy evaluation and columnar storage reduce memory usage
- **Familiar syntax**: Similar to tidyverse-style pipelines, making it accessible to R users
- **Type safety**: Strong typing system that catches errors early in development

The creators of Polars have provided exceptional [documentation](#) and [tutorials](#) that serve as valuable resources for learning and mastering the functionalities of the library.

Furthermore, several books are available that serve as introductions to Polars:

- [Python Polars: The Definitive Guide](#)

**i** Note

In this book, we assume that the reader has some experience with data manipulation concepts. This prior knowledge enables a more efficient and focused exploration of the clinical reporting concepts presented throughout the book.

To illustrate the basic usage of Polars, let's work with a sample ADSL dataset. This dataset contains subject-level information from a clinical trial, which will serve as a practical example for generating summaries using Polars.

```
import polars as pl

polars.config.Config

# Read clinical data
adsl = pl.read_parquet("data/adsl.parquet")

# Select columns
adsl = adsl.select(["USUBJID", "TRT01A", "AGE", "SEX"])

# Basic data exploration
adsl
```

| USUBJID       | TRT01A                 | AGE  | SEX      |
|---------------|------------------------|------|----------|
| str           | str                    | f64  | str      |
| "01-701-1015" | "Placebo"              | 63.0 | "Female" |
| "01-701-1023" | "Placebo"              | 64.0 | "Male"   |
| "01-701-1028" | "Xanomeline High Dose" | 71.0 | "Male"   |
| ...           | ...                    | ...  | ...      |
| "01-718-1371" | "Xanomeline High Dose" | 69.0 | "Female" |
| "01-718-1427" | "Xanomeline High Dose" | 74.0 | "Female" |

Key Polars operations for clinical reporting include:



### 4.5.1 I/O

Polars supports multiple data formats for input and output (see the [I/O guide](#)). For clinical development, we recommend the `.parquet` format because tools in Python, R, and Julia can read and write it without conversion. The example below loads subject-level ADSL data with Polars.

```
import polars as pl

adsl = pl.read_parquet("data/adsl.parquet")
adsl = adsl.select("STUDYID", "USUBJID", "TRT01A", "AGE", "SEX") # select columns
adsl
```

| STUDYID         | USUBJID       | TRT01A                 | AGE  | SEX      |
|-----------------|---------------|------------------------|------|----------|
| str             | str           | str                    | f64  | str      |
| "CDISCPILLOT01" | "01-701-1015" | "Placebo"              | 63.0 | "Female" |
| "CDISCPILLOT01" | "01-701-1023" | "Placebo"              | 64.0 | "Male"   |
| "CDISCPILLOT01" | "01-701-1028" | "Xanomeline High Dose" | 71.0 | "Male"   |
| ...             | ...           | ...                    | ...  | ...      |
| "CDISCPILLOT01" | "01-718-1371" | "Xanomeline High Dose" | 69.0 | "Female" |
| "CDISCPILLOT01" | "01-718-1427" | "Xanomeline High Dose" | 74.0 | "Female" |

### 4.5.2 Filtering

Filtering in Polars uses the `.filter()` method with column expressions. Below are examples applied to the ADSL data.

```
# Filter female subjects
adsl.filter(pl.col("SEX") == "Female")
```

| STUDYID         | USUBJID       | TRT01A                 | AGE  | SEX      |
|-----------------|---------------|------------------------|------|----------|
| str             | str           | str                    | f64  | str      |
| "CDISCPILLOT01" | "01-701-1015" | "Placebo"              | 63.0 | "Female" |
| "CDISCPILLOT01" | "01-701-1034" | "Xanomeline High Dose" | 77.0 | "Female" |
| "CDISCPILLOT01" | "01-701-1047" | "Placebo"              | 85.0 | "Female" |
| ...             | ...           | ...                    | ...  | ...      |

| STUDYID         | USUBJID       | TRT01A                 | AGE  | SEX      |
|-----------------|---------------|------------------------|------|----------|
| str             | str           | str                    | f64  | str      |
| "CDISCPILLOT01" | "01-718-1371" | "Xanomeline High Dose" | 69.0 | "Female" |
| "CDISCPILLOT01" | "01-718-1427" | "Xanomeline High Dose" | 74.0 | "Female" |

```
# Filter subjects with Age >= 65
adsl.filter(pl.col("AGE") >= 65)
```

| STUDYID         | USUBJID       | TRT01A                 | AGE  | SEX      |
|-----------------|---------------|------------------------|------|----------|
| str             | str           | str                    | f64  | str      |
| "CDISCPILLOT01" | "01-701-1028" | "Xanomeline High Dose" | 71.0 | "Male"   |
| "CDISCPILLOT01" | "01-701-1033" | "Xanomeline Low Dose"  | 74.0 | "Male"   |
| "CDISCPILLOT01" | "01-701-1034" | "Xanomeline High Dose" | 77.0 | "Female" |
| ...             | ...           | ...                    | ...  | ...      |
| "CDISCPILLOT01" | "01-718-1371" | "Xanomeline High Dose" | 69.0 | "Female" |
| "CDISCPILLOT01" | "01-718-1427" | "Xanomeline High Dose" | 74.0 | "Female" |

### 4.5.3 Deriving

Deriving new variables is common in clinical data analysis for creating age groups, BMI categories, or treatment flags. Polars uses `.with_columns()` to add new columns while keeping existing ones.

```
# Create age groups
adsl.with_columns([
    pl.when(pl.col("AGE") < 65)
        .then(pl.lit("<65"))
        .otherwise(pl.lit(">=65"))
        .alias("AGECAT")
])
```

| STUDYID         | USUBJID       | TRT01A    | AGE  | SEX      | AGECAT |
|-----------------|---------------|-----------|------|----------|--------|
| str             | str           | str       | f64  | str      | str    |
| "CDISCPILLOT01" | "01-701-1015" | "Placebo" | 63.0 | "Female" | "<65"  |
| "CDISCPILLOT01" | "01-701-1023" | "Placebo" | 64.0 | "Male"   | "<65"  |

| STUDYID         | USUBJID       | TRT01A                 | AGE  | SEX      | AGECAT |
|-----------------|---------------|------------------------|------|----------|--------|
| str             | str           | str                    | f64  | str      | str    |
| "CDISCPILLOT01" | "01-701-1028" | "Xanomeline High Dose" | 71.0 | "Male"   | ">=65" |
| ...             | ...           | ...                    | ...  | ...      | ...    |
| "CDISCPILLOT01" | "01-718-1371" | "Xanomeline High Dose" | 69.0 | "Female" | ">=65" |
| "CDISCPILLOT01" | "01-718-1427" | "Xanomeline High Dose" | 74.0 | "Female" | ">=65" |

#### 4.5.4 Grouping

Grouping operations are fundamental for creating summary statistics in clinical reports. Polars uses `group_by()` followed by aggregation functions to compute counts, means, and other statistics by categorical variables like treatment groups.

The `.count()` method provides a quick way to get subject counts by group.

```
# Count by treatment group
adsl.group_by("TRT01A").len().sort("TRT01A")
```

| TRT01A                 | len |
|------------------------|-----|
| str                    | u32 |
| "Placebo"              | 86  |
| "Xanomeline High Dose" | 84  |
| "Xanomeline Low Dose"  | 84  |

You can also use `.agg()` with multiple aggregation functions:

```
# Age statistics by treatment group
adsl.group_by("TRT01A").agg([
    pl.col("AGE").mean().round(1).alias("mean_age"),
    pl.col("AGE").std().round(2).alias("sd_age")
]).sort("TRT01A")
```

| TRT01A                 | mean_age | sd_age |
|------------------------|----------|--------|
| str                    | f64      | f64    |
| "Placebo"              | 75.2     | 8.59   |
| "Xanomeline High Dose" | 74.4     | 7.89   |
| "Xanomeline Low Dose"  | 75.7     | 8.29   |

### 4.5.5 Joining

Joining datasets is essential for combining subject-level data (ADSL) with event-level data (e.g. ADAE, ADLB). Polars supports various join types including inner, left, and full joins.

Here is a toy example that splits ADSL and joins it back by USUBJID.

```
# Create a simple demographics subset
demo = adsl.select("USUBJID", "AGE", "SEX").head(3)

# Create treatment info subset
trt = adsl.select("USUBJID", "TRT01A").head(3)

# Left join to combine datasets
demo.join(trt, on="USUBJID", how="left")
```

| USUBJID       | AGE  | SEX      | TRT01A                 |
|---------------|------|----------|------------------------|
| str           | f64  | str      | str                    |
| "01-701-1015" | 63.0 | "Female" | "Placebo"              |
| "01-701-1023" | 64.0 | "Male"   | "Placebo"              |
| "01-701-1028" | 71.0 | "Male"   | "Xanomeline High Dose" |

### 4.5.6 Pivoting

Pivoting transforms data from long to wide format, commonly needed for creating tables. Use `.pivot()` to reshape grouped data into columns.

```
# Create summary by treatment and sex
(
    adsl
        .group_by(["TRT01A", "SEX"])
        .agg(pl.len().alias("n"))
        .pivot(
            values="n",
            index="SEX",
            on="TRT01A"
        )
)
```

| SEX      | Xanomeline Low Dose | Placebo | Xanomeline High Dose |
|----------|---------------------|---------|----------------------|
| str      | u32                 | u32     | u32                  |
| "Male"   | 34                  | 33      | 44                   |
| "Female" | 50                  | 53      | 40                   |

Having covered the essential Polars operations for data manipulation, we now turn to the second component of our clinical reporting workflow: formatting and presenting the processed data in regulatory-compliant RTF format.

## 4.6 rtflite

```
import rtflite as rtf
```

rtflite is a Python package for creating production-ready tables and figures in RTF format. While Polars handles the data processing and statistical calculations, rtflite focuses exclusively on the presentation layer. The package is designed to:

- Provide simple Python classes that map to table elements (title, headers, body, footnotes) for intuitive table construction.
- Offer a canonical Python API with a clear, composable interface.

- Focus exclusively on **table formatting and layout**, leaving data manipulation to dataframe libraries like polars or pandas.
- Minimize external dependencies for maximum portability and reliability.

Creating an RTF table involves three steps:

- Design the desired table layout and structure.
- Configure the appropriate rtfite components.
- Generate and save the RTF document.

This guide introduces rtfite’s core components and demonstrates how to turn dataframes into Tables, Listings, and Figures (TLFs) for clinical reporting.

#### 4.6.1 Data: adverse events

To explore the RTF generation capabilities in rtfite, we will use the dataset `data/adae.parquet`. This dataset contains adverse event (AE) information from a clinical trial.

Below are the meanings of relevant variables:

- USUBJID: Unique Subject Identifier
- TRTA: Actual Treatment
- AEDECOD: Dictionary-Derived Term

```
# Load adverse events data
df = pl.read_parquet("data/adae.parquet")

df.select(["USUBJID", "TRTA", "AEDECOD"])
```

| USUBJID       | TRTA                   | AEDECOD                     |
|---------------|------------------------|-----------------------------|
| str           | str                    | str                         |
| "01-701-1015" | "Placebo"              | "APPLICATION SITE ERYTHEMA" |
| "01-701-1015" | "Placebo"              | "APPLICATION SITE PRURITUS" |
| "01-701-1015" | "Placebo"              | "DIARRHOEA"                 |
| ...           | ...                    | ...                         |
| "01-718-1427" | "Xanomeline High Dose" | "DECREASED APPETITE"        |

| USUBJID       | TRTA                   | AEDECOD  |
|---------------|------------------------|----------|
| str           | str                    | str      |
| "01-718-1427" | "Xanomeline High Dose" | "NAUSEA" |

### 4.6.2 Table-ready data

In this AE example, we provide the number of subjects with each type of AE by treatment group.

```
tbl = (
  df.group_by(["TRTA", "AEDECOD"])
    .agg(pl.len().alias("n"))
    .sort("TRTA")
    .pivot(values="n", index="AEDECOD", on="TRTA")
    .fill_null(0)
    .sort("AEDECOD") # Sort by adverse event name to match R output
)

tbl
```

| AEDECOD                | Placebo | Xanomeline High Dose | Xanomeline Low Dose |
|------------------------|---------|----------------------|---------------------|
| str                    | u32     | u32                  | u32                 |
| "ABDOMINAL DISCOMFORT" | 0       | 1                    | 0                   |
| "ABDOMINAL PAIN"       | 1       | 2                    | 3                   |
| "ACROCHORDON EXCISION" | 0       | 1                    | 0                   |
| ...                    | ...     | ...                  | ...                 |
| "WOUND"                | 0       | 0                    | 2                   |
| "WOUND HAEMORRHAGE"    | 0       | 1                    | 0                   |

### 4.6.3 Table component classes

rtflite provides dedicated classes for each table component. Commonly used classes include:

- **RTFPage:** RTF page information (orientation, margins, pagination).

- **RTFPageHeader**: Page headers with page numbering (compatible with r2rtf).
- **RTFPageFooter**: Page footers for attribution and notices.
- **RTFTitle**: RTF title information.
- **RTFColumnHeader**: RTF column header information.
- **RTFBody**: RTF table body information.
- **RTFFootnote**: RTF footnote information.
- **RTFSource**: RTF data source information.

These component classes work together to build complete RTF documents. A full list of all classes and their parameters can be found in the [API reference](#).

#### 4.6.4 Simple example

A minimal example below illustrates how to combine components to create an RTF table.

- `RTFBody()` defines table body layout.
- `RTFDocument()` transfers table layout information into RTF syntax.
- `write_rtf()` saves encoded RTF into a `.rtf` file.

```
# Create simple RTF document
doc = rtf.RTFDocument(
    df=tbl.head(6),
    rtf_body=rtf.RTFBody(), # Step 1: Add table attributes
)

# Step 2 & 3: Convert to RTF and write to file
doc.write_rtf("rtf/tlf_overview1.rtf")
```

```
rtf/tlf_overview1.rtf
```

```
PosixPath('pdf/tlf_overview1.pdf')
```



### 4.6.5 Column width

If we want to adjust the width of each column to provide more space to the first column, this can be achieved by updating `col_rel_width` in `RTFBody`.

The input of `col_rel_width` is a list with the same length as the number of columns. This argument defines the relative length of each column within a pre-defined total column width.

In this example, the defined relative width is 3:2:2:2. Only the ratio of `col_rel_width` is used. Therefore it is equivalent to use `col_rel_width = [6,4,4,4]` or `col_rel_width = [1.5,1,1,1]`.

```
# Create RTF document with custom column widths
doc = rtf.RTFDocument(
    df=tbl.head(6),
    rtf_body=rtf.RTFBody(
        col_rel_width=[3, 2, 2, 2] # Define relative width
    ),
)

doc.write_rtf("rtf/tlf_overview2.rtf")
```

rtf/tlf\_overview2.rtf

PosixPath('pdf/tlf\_overview2.pdf')

### 4.6.6 Column headers

In `RTFColumnHeader`, the `text` argument provides the column header content as a list of strings.

```
# Create RTF document with column headers
doc = rtf.RTFDocument(
    df=tbl.head(6),
    rtf_column_header=rtf.RTFColumnHeader(
        text=[
            "Adverse Events",
```

```

        "Placebo",
        "Xanomeline High Dose",
        "Xanomeline Low Dose",
    ],
),
rtf_body=rtf.RTFBody(col_rel_width=[3, 2, 2, 2]),
)

doc.write_rtf("rtf/tlf_overview3.rtf")

```

rtf/tlf\_overview3.rtf

PosixPath('pdf/tlf\_overview3.pdf')

We also allow column headers to be displayed in multiple lines. If an empty column name is needed for a column, you can insert an empty string. For example, ["name 1", "", "name 3"].

In `RTFColumnHeader`, the `col_rel_width` can be used to align column headers with different numbers of columns.

By using `RTFColumnHeader` with `col_rel_width`, one can customize complex column headers. If there are multiple pages, the column header will repeat on each page by default.

```

# Create RTF document with multi-line column headers
doc = rtf.RTFDocument(
    df=tbl.head(50),
    rtf_page=rtf.RTFPage(nrow=15),
    rtf_column_header=[
        rtf.RTFColumnHeader(text=[" ", "Treatment"], col_rel_width=[3, 3]),
        rtf.RTFColumnHeader(
            text=[
                "Adverse Events",
                "Placebo",
                "Xanomeline High Dose",
                "Xanomeline Low Dose",
            ],
            col_rel_width=[3, 1, 1, 1],
        ),
    ],
)

```

```

],
  rtf_body=rtf.RTFBody(col_rel_width=[3, 1, 1, 1]),
)

doc.write_rtf("rtf/tlf_overview4.rtf")

```

```
rtf/tlf_overview4.rtf
```

```
PosixPath('pdf/tlf_overview4.pdf')
```

#### 4.6.7 Titles, footnotes, and data source

RTF documents can include additional components to provide context and documentation:

- RTFTitle: Add document titles and subtitles
- RTFFootnote: Add explanatory footnotes
- RTFSource: Add data source attribution

```

# Create RTF document with titles, footnotes, and source
doc = rtf.RTFDocument(
  df=tbl.head(15),
  rtf_title=rtf.RTFTitle(
    text=["Summary of Adverse Events by Treatment Group", "Safety Analysis Set"]
  ),
  rtf_column_header=rtf.RTFColumnHeader(
    text=[
      "Adverse Events",
      "Placebo\\line (N=86)",
      "Xanomeline High Dose\\line (N=84)",
      "Xanomeline Low Dose\\line (N=84)",
    ],
  ),
  rtf_body=rtf.RTFBody(col_rel_width=[3, 2, 2, 2]),
  rtf_footnote=rtf.RTFFootnote(
    text=[
      "Adverse events are coded using MedDRA version 25.0.",
      "Events are sorted alphabetically by preferred term.",
    ],
  )

```

```

    ),
    rtf_source=rtf.RTFSource(text="Source: ADAE dataset, Data cutoff: 01JAN2023"),
)

doc.write_rtf("rtf/tlf_overview5.rtf")

```

rtf/tlf\_overview5.rtf

PosixPath('pdf/tlf\_overview5.pdf')

Note the use of `\\line` in column headers to create line breaks within cells.

## 4.6.8 Text formatting and alignment

rtflite supports various text formatting options:

- **Text formatting:** Bold (`b`), italic (`i`), underline (`u`), strikethrough (`s`)
- **Text alignment:** Left (`l`), center (`c`), right (`r`), justify (`j`)
- **Font properties:** Font size, font family

```

# Create RTF document with text formatting and alignment
doc = rtf.RTFDocument(
    df=tbl.head(10),
    rtf_column_header=rtf.RTFColumnHeader(
        text=[
            "Adverse Events",
            "Placebo",
            "Xanomeline High Dose",
            "Xanomeline Low Dose",
        ],
        text_format="b", # Bold headers
        text_justification=["l", "c", "c", "c"],
    ),
    rtf_body=rtf.RTFBody(
        col_rel_width=[3, 1, 1, 1],

```

```

        text_justification=["l", "c", "c", "c"],
    ),
)

doc.write_rtf("rtf/tlf_overview6.rtf")

```

rtf/tlf\_overview6.rtf

PosixPath('pdf/tlf\_overview6.pdf')

#### 4.6.9 Border customization

Table borders can be customized extensively:

- **Border styles:** single, double, thick, dotted, dashed
- **Border sides:** border\_top, border\_bottom, border\_left, border\_right
- **Page borders:** border\_first, border\_last for first/last rows across pages

```

# Create RTF document with custom borders
doc = rtf.RTFDocument(
    df=tbl.head(8),
    rtf_column_header=rtf.RTFColumnHeader(
        text=[
            "Adverse Events",
            "Placebo",
            "Xanomeline High Dose",
            "Xanomeline Low Dose",
        ],
        border_bottom=["single", "double", "single", "single"],
    ),
    rtf_body=rtf.RTFBody(
        col_rel_width=[3, 2, 2, 2],
        border_left=["single", "", "", ""],
    ),
)

doc.write_rtf("rtf/tlf_overview7.rtf")

```

```
rtf/tlf_overview7.rtf
```

```
PosixPath('pdf/tlf_overview7.pdf')
```

## 4.7 Next steps

Having covered the fundamental concepts and tools for creating clinical TLFs with Python, readers can explore specific implementations based on their requirements:

Each chapter provides step-by-step tutorials with reproducible code examples that can be adapted for specific clinical reporting requirements.

## 5 Disposition of participants

### Objective

Create participant disposition tables to track how participants flow through the study from enrollment to completion. Learn to analyze completion status and discontinuation reasons using Polars and create regulatory-compliant disposition tables with `rtflite`.

### 5.1 Overview

Clinical trials need to track how participants flow through a study from enrollment to completion. Following [ICH E3 guidance](#), regulatory submissions require a disposition table in Section 10.1 that summarizes:

- **Enrolled:** Total participants who entered the study
- **Completed:** Participants who finished the study protocol
- **Discontinued:** Participants who left early and their reasons

This tutorial shows you how to create a regulatory-compliant disposition table using Python's `rtflite` package.

```
import polars as pl # Manipulate data
import rtflite as rtf # Reporting in RTF format
```

```
polars.config.Config
```

## 5.2 Step 1: Load data

We start by loading the Subject-level Analysis Dataset (ADSL), which contains all participant information needed for our disposition table.

The ADSL dataset stores participant-level information including treatment assignments and study completion status. We're using the `parquet` format for data storage.

```
adsl = pl.read_parquet("data/adsl.parquet")
```

Let's examine the key variables we'll use to build our disposition table:

- **USUBJID**: Unique identifier for each participant
- **TRT01P**: Treatment name (text)
- **TRT01PN**: Treatment group (numeric code)
- **DISCONFL**: Flag indicating if participant discontinued (Y/N)
- **DCREASCD**: Specific reason for discontinuation

```
adsl.select(["USUBJID", "TRT01P", "TRT01PN", "DISCONFL", "DCREASCD"])
```

| USUBJID       | TRT01P                 | TRT01PN | DISCONFL | DCREASCD           |
|---------------|------------------------|---------|----------|--------------------|
| str           | str                    | i64     | str      | str                |
| "01-701-1015" | "Placebo"              | 0       | " "      | "Completed"        |
| "01-701-1023" | "Placebo"              | 0       | "Y"      | "Adverse Event"    |
| "01-701-1028" | "Xanomeline High Dose" | 81      | " "      | "Completed"        |
| ...           | ...                    | ...     | ...      | ...                |
| "01-718-1371" | "Xanomeline High Dose" | 81      | "Y"      | "Adverse Event"    |
| "01-718-1427" | "Xanomeline High Dose" | 81      | "Y"      | "Lack of Efficacy" |

## 5.3 Step 2: Count total participants

First, we count how many participants were enrolled in each treatment group.



We group participants by treatment arm and count them using `.group_by()` and `.agg()`. The `.pivot()` operation reshapes our data from long format (rows for each treatment) to wide format (columns for each treatment), which matches the standard disposition table layout.

```
n_rand = (
    adsl
    .group_by("TRT01PN")
    .agg(n = pl.len())
    .with_columns([
        pl.lit("Participants in population").alias("row"),
        pl.lit(None, dtype=pl.Float64).alias("pct") # Placeholder for percentage (not applicable)
    ])
    .pivot(
        index="row",
        on="TRT01PN",
        values=["n", "pct"],
        sort_columns=True
    )
)

n_rand
```

| row                          | n_0 | n_54 | n_81 | pct_0 | pct_54 | pct_81 |
|------------------------------|-----|------|------|-------|--------|--------|
| str                          | u32 | u32  | u32  | f64   | f64    | f64    |
| "Participants in population" | 86  | 84   | 84   | null  | null   | null   |

## 5.4 Step 3: Count completed participants

Next, we identify participants who successfully completed the study and calculate what percentage they represent of each treatment group.

We filter for participants where `DCREASCD == "Completed"`, then calculate both counts and percentages. The `.join()` operation brings in the total count for each treatment group so we can compute percentages.

```

n_complete = (
    adsl
    .filter(pl.col("DCREASCD") == "Completed")
    .group_by("TRT01PN")
    .agg(n = pl.len())
    .join(
        adsl.group_by("TRT01PN").agg(total = pl.len()),
        on="TRT01PN"
    )
    .with_columns([
        pl.lit("Completed").alias("row"),
        (100.0 * pl.col("n") / pl.col("total")).round(1).alias("pct")
    ])
    .pivot(
        index="row",
        on="TRT01PN",
        values=["n", "pct"],
        sort_columns=True
    )
)

n_complete

```

| row         | n_0 | n_54 | n_81 | pct_0 | pct_54 | pct_81 |
|-------------|-----|------|------|-------|--------|--------|
| str         | u32 | u32  | u32  | f64   | f64    | f64    |
| "Completed" | 58  | 25   | 27   | 67.4  | 29.8   | 32.1   |

## 5.5 Step 4: Count discontinued participants

Now we count participants who left the study early, regardless of their specific reason.

We filter for participants where the discontinuation flag `DISCONFL == "Y"`, then follow the same pattern of counting and calculating percentages within each treatment group.

```

n_disc = (
    adsl
    .filter(pl.col("DISCONFL") == "Y")
    .group_by("TRT01PN")
    .agg(n = pl.len())
    .join(
        adsl.group_by("TRT01PN").agg(total = pl.len()),
        on="TRT01PN"
    )
    .with_columns([
        pl.lit("Discontinued").alias("row"),
        (100.0 * pl.col("n") / pl.col("total")).round(1).alias("pct")
    ])
    .pivot(
        index="row",
        on="TRT01PN",
        values=["n", "pct"],
        sort_columns=True
    )
)

n_disc

```

| row            | n_0 | n_54 | n_81 | pct_0 | pct_54 | pct_81 |
|----------------|-----|------|------|-------|--------|--------|
| str            | u32 | u32  | u32  | f64   | f64    | f64    |
| "Discontinued" | 28  | 59   | 57   | 32.6  | 70.2   | 67.9   |

## 5.6 Step 5: Break down discontinuation reasons

For regulatory reporting, we need to show the specific reasons why participants discontinued.

We filter out completed participants, then group by both treatment and discontinuation reason. The indentation (four spaces) in the row labels helps show these are subcategories under “Discontinued”. We also use `.fill_null(0)` to handle cases where

certain discontinuation reasons don't occur in all treatment groups.

```
n_reason = (
  adsl
  .filter(pl.col("DCREASCD") != "Completed")
  .group_by(["TRT01PN", "DCREASCD"])
  .agg(n = pl.len())
  .join(
    adsl.group_by("TRT01PN").agg(total = pl.len()),
    on="TRT01PN"
  )
  .with_columns([
    pl.concat_str([pl.lit("    "), pl.col("DCREASCD")]).alias("row"),
    (100.0 * pl.col("n") / pl.col("total")).round(1).alias("pct")
  ])
  .pivot(
    index="row",
    on="TRT01PN",
    values=["n", "pct"],
    sort_columns=True
  )
  .with_columns([
    pl.col(["n_0", "n_54", "n_81"]).fill_null(0),
    pl.col(["pct_0", "pct_54", "pct_81"]).fill_null(0.0)
  ])
  .sort("row")
)

n_reason
```

| row |                   | n_0 | n_54 | n_81 | pct_0 | pct_54 | pct_81 |
|-----|-------------------|-----|------|------|-------|--------|--------|
| str |                   | u32 | u32  | u32  | f64   | f64    | f64    |
| "   | Adverse Event"    | 8   | 44   | 40   | 9.3   | 52.4   | 47.6   |
| "   | Death"            | 2   | 1    | 0    | 2.3   | 1.2    | 0.0    |
| "   | I/E Not Met"      | 1   | 0    | 2    | 1.2   | 0.0    | 2.4    |
| ... |                   | ... | ...  | ...  | ...   | ...    | ...    |
| "   | Sponsor Decision" | 2   | 2    | 3    | 2.3   | 2.4    | 3.6    |
| "   | Withdrew Consent" | 9   | 10   | 8    | 10.5  | 11.9   | 9.5    |

## 5.7 Step 6: Combine all results

Now we stack all our individual summaries together to create the complete disposition table.

Using `pl.concat()`, we combine the enrollment counts, completion counts, discontinuation counts, and detailed discontinuation reasons into a single table that flows logically from top to bottom.

```
tbl_disp = pl.concat([
    n_rand,
    n_complete,
    n_disc,
    n_reason
])

tbl_disp
```

| row                          | n_0 | n_54 | n_81 | pct_0 | pct_54 | pct_81 |
|------------------------------|-----|------|------|-------|--------|--------|
| str                          | u32 | u32  | u32  | f64   | f64    | f64    |
| "Participants in population" | 86  | 84   | 84   | null  | null   | null   |
| "Completed"                  | 58  | 25   | 27   | 67.4  | 29.8   | 32.1   |
| "Discontinued"               | 28  | 59   | 57   | 32.6  | 70.2   | 67.9   |
| ...                          | ... | ...  | ...  | ...   | ...    | ...    |
| " Sponsor Decision"          | 2   | 2    | 3    | 2.3   | 2.4    | 3.6    |
| " Withdrew Consent"          | 9   | 10   | 8    | 10.5  | 11.9   | 9.5    |

## 5.8 Step 7: Generate publication-ready output

Finally, we format our table in RTF format using the `rtflite` package.

The `RTFDocument` class handles the complex formatting required for clinical reports, including proper column headers, borders, and spacing. The resulting RTF file can be directly included in regulatory submissions or converted to PDF for review.

```

doc_disp = rtf.RTFDocument(
  df=tbl_disp.select("row", "n_0", "pct_0", "n_54", "pct_54", "n_81", "pct_81"),
  rtf_title=rtf.RTFTitle(text=["Disposition of Participants"]),
  rtf_column_header=[
    rtf.RTFColumnHeader(
      text=["", "Placebo", "Xanomeline Low Dose", "Xanomeline High Dose"],
      col_rel_width=[3] + [2] * 3,
      text_justification=["l"] + ["c"] * 3,
    ),
    rtf.RTFColumnHeader(
      text=["", "n", "(%)", "n", "(%)", "n", "(%)"],
      col_rel_width=[3] + [1] * 6,
      text_justification=["l"] + ["c"] * 6,
      border_top=[""] + ["single"] * 6,
      border_left=["single"] + ["single", ""] * 3
    )
  ],
  rtf_body=rtf.RTFBody(
    col_rel_width=[3] + [1] * 6,
    text_justification=["l"] + ["c"] * 6,
    border_left=["single"] + ["single", ""] * 3
  ),
  rtf_source=rtf.RTFSource(text=["Source: ADSL dataset"]) # Required source attribution
)

doc_disp.write_rtf("rtf/tlf_disposition.rtf") # Save as RTF for submission

```

```
rtf/tlf_disposition.rtf
```

```
PosixPath('pdf/tlf_disposition.pdf')
```

## 6 Study population

### Objective

Create study population summary tables to document participant counts across different analysis populations. Learn to use population flags in ADSL data and generate regulatory-compliant population tables with `rtflite`.

### 6.1 Overview

Clinical trials define multiple analysis populations based on different inclusion criteria. Following [ICH E3 guidance](#), regulatory submissions must clearly document the number of participants in each analysis population to support the validity of statistical analyses.

The key analysis populations typically include:

- **All Randomized:** Total participants who entered the study
- **Intent-to-Treat (ITT):** Participants included in the primary efficacy analysis
- **Efficacy Population:** Participants who meet specific criteria for efficacy evaluation
- **Safety Population:** Participants who received at least one dose of study treatment

This tutorial shows you how to create a population summary table using Python's `rtflite` package.

```
import polars as pl # Data manipulation
import rtflite as rtf # RTF reporting
```

```
polars.config.Config
```

## 6.2 Step 1: Load data

We start by loading the Subject-level Analysis Dataset (ADSL), which contains population flags for each participant.

```
adsl = pl.read_parquet("data/adsl.parquet")
```

Let's examine the key population flag variables we'll use:

- **USUBJID**: Unique participant identifier
- **TRT01P**: Planned treatment group
- **ITTFL**: Intent-to-treat population flag (Y/N)
- **EFFFL**: Efficacy population flag (Y/N)
- **SAFFL**: Safety population flag (Y/N)

```
adsl.select(["USUBJID", "TRT01P", "ITTFL", "EFFFL", "SAFFL"])
```

| USUBJID       | TRT01P                 | ITTFL | EFFFL | SAFFL |
|---------------|------------------------|-------|-------|-------|
| str           | str                    | str   | str   | str   |
| "01-701-1015" | "Placebo"              | "Y"   | "Y"   | "Y"   |
| "01-701-1023" | "Placebo"              | "Y"   | "Y"   | "Y"   |
| "01-701-1028" | "Xanomeline High Dose" | "Y"   | "Y"   | "Y"   |
| ...           | ...                    | ...   | ...   | ...   |
| "01-718-1371" | "Xanomeline High Dose" | "Y"   | "Y"   | "Y"   |
| "01-718-1427" | "Xanomeline High Dose" | "Y"   | "Y"   | "Y"   |

## 6.3 Step 2: Calculate treatment group totals

First, we calculate the total number of randomized participants in each treatment group, which will serve as the denominator for percentage calculations.



```
totals = adsl.group_by("TRT01P").agg(
    total = pl.len()
)

totals
```

| TRT01P                 | total |
|------------------------|-------|
| str                    | u32   |
| "Placebo"              | 86    |
| "Xanomeline High Dose" | 84    |
| "Xanomeline Low Dose"  | 84    |

## 6.4 Step 3: Define helper function

We create a reusable function to count participants by treatment group for any population subset.

```
def count_by_treatment(data, population_name):
    """Count participants by treatment group and add population label"""
    return data.group_by("TRT01P").agg(
        n = pl.len()
    ).with_columns(
        population = pl.lit(population_name)
    )
```

## 6.5 Step 4: Count each population

Now we calculate participant counts for each analysis population.

### 6.5.1 All randomized participants

```
pop_all = count_by_treatment(
    data=adsl,
    population_name="Participants in population"
)

pop_all
```

| TRT01P                 | n   | population                   |
|------------------------|-----|------------------------------|
| str                    | u32 | str                          |
| "Xanomeline High Dose" | 84  | "Participants in population" |
| "Placebo"              | 86  | "Participants in population" |
| "Xanomeline Low Dose"  | 84  | "Participants in population" |

### 6.5.2 Intent-to-treat population

```
adsl_itt = adsl.filter(pl.col("ITTFL") == "Y")
pop_itt = count_by_treatment(
    data=adsl_itt,
    population_name="Participants included in ITT population"
)

pop_itt
```

| TRT01P                 | n   | population                          |
|------------------------|-----|-------------------------------------|
| str                    | u32 | str                                 |
| "Placebo"              | 86  | "Participants included in ITT p..." |
| "Xanomeline Low Dose"  | 84  | "Participants included in ITT p..." |
| "Xanomeline High Dose" | 84  | "Participants included in ITT p..." |

### 6.5.3 Efficacy population

```
adsl_eff = adsl.filter(pl.col("EFFFL") == "Y")
pop_eff = count_by_treatment(
```

```

    data=adsl_eff,
    population_name="Participants included in efficacy population"
)

pop_eff

```

| TRT01P                 | n   | population                         |
|------------------------|-----|------------------------------------|
| str                    | u32 | str                                |
| "Xanomeline Low Dose"  | 81  | "Participants included in effic... |
| "Placebo"              | 79  | "Participants included in effic... |
| "Xanomeline High Dose" | 74  | "Participants included in effic... |

### 6.5.4 Safety population

```

adsl_saf = adsl.filter(pl.col("SAFFL") == "Y")
pop_saf = count_by_treatment(
    data=adsl_saf,
    population_name="Participants included in safety population"
)

pop_saf

```

| TRT01P                 | n   | population                         |
|------------------------|-----|------------------------------------|
| str                    | u32 | str                                |
| "Xanomeline High Dose" | 84  | "Participants included in safet... |
| "Xanomeline Low Dose"  | 84  | "Participants included in safet... |
| "Placebo"              | 86  | "Participants included in safet... |

## 6.6 Step 5: Combine all populations

We stack all population counts together into a single dataset.

```
all_populations = pl.concat([
    pop_all,
    pop_itt,
    pop_eff,
    pop_saf
])

all_populations
```

| TRT01P                 | n   | population                          |
|------------------------|-----|-------------------------------------|
| str                    | u32 | str                                 |
| "Xanomeline High Dose" | 84  | "Participants in population"        |
| "Placebo"              | 86  | "Participants in population"        |
| "Xanomeline Low Dose"  | 84  | "Participants in population"        |
| ...                    | ... | ...                                 |
| "Xanomeline Low Dose"  | 84  | "Participants included in safet..." |
| "Placebo"              | 86  | "Participants included in safet..." |

## 6.7 Step 6: Calculate percentages

We join with the total counts and calculate what percentage each population represents of the total randomized participants.

```
stats_with_pct = all_populations.join(
    totals,
    on="TRT01P"
).with_columns(
    pct = (100.0 * pl.col("n") / pl.col("total")).round(1)
)

stats_with_pct
```

| TRT01P                 | n   | population                   | total | pct   |
|------------------------|-----|------------------------------|-------|-------|
| str                    | u32 | str                          | u32   | f64   |
| "Xanomeline High Dose" | 84  | "Participants in population" | 84    | 100.0 |
| "Placebo"              | 86  | "Participants in population" | 86    | 100.0 |

| TRT01P<br>str         | n<br>u32 | population<br>str                  | total<br>u32 | pct<br>f64 |
|-----------------------|----------|------------------------------------|--------------|------------|
| "Xanomeline Low Dose" | 84       | "Participants in population"       | 84           | 100.0      |
| ...                   | ...      | ...                                | ...          | ...        |
| "Xanomeline Low Dose" | 84       | "Participants included in safet... | 84           | 100.0      |
| "Placebo"             | 86       | "Participants included in safet... | 86           | 100.0      |

## 6.8 Step 7: Format display values

For the final table, we format the display text. The total randomized count shows just “N”, while subset populations show “N (%)”.

```
formatted_stats = stats_with_pct.with_columns(
    display = pl.when(pl.col("population") == "Participants in population")
        .then(pl.col("n").cast(str))
        .otherwise(
            pl.concat_str([
                pl.col("n").cast(str),
                pl.lit(" ("),
                pl.col("pct").round(1).cast(str),
                pl.lit(")")
            ])
        )
)

formatted_stats
```

| TRT01P<br>str          | n<br>u32 | population<br>str                  | total<br>u32 | pct<br>f64 | display<br>str |
|------------------------|----------|------------------------------------|--------------|------------|----------------|
| "Xanomeline High Dose" | 84       | "Participants in population"       | 84           | 100.0      | "84"           |
| "Placebo"              | 86       | "Participants in population"       | 86           | 100.0      | "86"           |
| "Xanomeline Low Dose"  | 84       | "Participants in population"       | 84           | 100.0      | "84"           |
| ...                    | ...      | ...                                | ...          | ...        | ...            |
| "Xanomeline Low Dose"  | 84       | "Participants included in safet... | 84           | 100.0      | "84 (100.0)"   |
| "Placebo"              | 86       | "Participants included in safet... | 86           | 100.0      | "86 (100.0)"   |

## 6.9 Step 8: Create final table

We reshape the data from long format (rows for each treatment-population combination) to wide format (columns for each treatment group).

```
df_overview = formatted_stats.pivot(  
    values="display",  
    index="population",  
    on="TRT01P",  
    maintain_order=True  
)<div data-bbox="111 424 889 543" data-label="Table">

| population                         | Placebo      | Xanomeline Low Dose | Xanomeline High Dose |
|------------------------------------|--------------|---------------------|----------------------|
| str                                | str          | str                 | str                  |
| "Participants in population"       | "86"         | "84"                | "84"                 |
| "Participants included in ITT p... | "86 (100.0)" | "84 (100.0)"        | "84 (100.0)"         |
| "Participants included in effic... | "79 (91.9)"  | "81 (96.4)"         | "74 (88.1)"          |
| "Participants included in safet... | "86 (100.0)" | "84 (100.0)"        | "84 (100.0)"         |


```

## 6.10 Step 9: Generate publication-ready output

Finally, we format the population table for regulatory submission using the `rtflite` package.

```
doc_overview = rtf.RTFDocument(  
    df=df_overview,  
    rtf_title=rtf.RTFTitle(  
        text=["Analysis Population", "All Participants Randomized"]  
    ),  
    rtf_column_header=rtf.RTFColumnHeader(  
        text=["", "Placebo\\nn (%)", "Xanomeline Low Dose\\nn (%)", "Xanomeline High Dose\\nn (%)"]  
    )  
)
```

```

        col_rel_width=[4, 2, 2, 2],
        text_justification=["l", "c", "c", "c"],
    ),
    rtf_body=rtf.RTFBody(
        col_rel_width=[4, 2, 2, 2],
        text_justification=["l", "c", "c", "c"],
    ),
    rtf_source=rtf.RTFSource(text=["Source: ADSL dataset"])
)

doc_overview.write_rtf("rtf/tlf_population.rtf")

```

rtf/tlf\_population.rtf

PosixPath('pdf/tlf\_population.pdf')

## 7 Baseline characteristics

### Objective

Create baseline characteristics tables to summarize demographic and clinical characteristics of study participants at enrollment. Learn to calculate descriptive statistics by treatment group using Polars and format regulatory-compliant tables with `rtflite`.

### 7.1 Overview

Baseline characteristics tables summarize the demographic and clinical characteristics of study participants at enrollment. Following [ICH E3 guidance](#), these tables are essential for understanding the study population and assessing comparability between treatment groups.

This tutorial shows you how to create a baseline characteristics table using Python's `rtflite` package.

```
import polars as pl # Data manipulation
import rtflite as rtf # RTF reporting
```

```
polars.config.Config
```

### 7.2 Step 1: Load data

We start by loading the Subject-level Analysis Dataset (ADSL) and filtering to the safety population.



```

adsl = (
    pl.read_parquet("data/adsl.parquet")
    .select(["USUBJID", "TRT01P", "AGE", "SEX", "RACE"])
)

adsl

```

| USUBJID       | TRT01P                 | AGE  | SEX      | RACE                        |
|---------------|------------------------|------|----------|-----------------------------|
| str           | str                    | f64  | str      | str                         |
| "01-701-1015" | "Placebo"              | 63.0 | "Female" | "White"                     |
| "01-701-1023" | "Placebo"              | 64.0 | "Male"   | "White"                     |
| "01-701-1028" | "Xanomeline High Dose" | 71.0 | "Male"   | "White"                     |
| ...           | ...                    | ...  | ...      | ...                         |
| "01-718-1371" | "Xanomeline High Dose" | 69.0 | "Female" | "White"                     |
| "01-718-1427" | "Xanomeline High Dose" | 74.0 | "Female" | "Black Or African American" |

## 7.3 Step 2: Calculate summary statistics

We'll create separate functions to handle continuous and categorical variables.

### 7.3.1 Continuous variables (age)

For continuous variables, we calculate mean (SD) and median [min, max].

```

def summarize_continuous(df, var):
    """Calculate summary statistics for continuous variables"""
    return df.group_by("TRT01P").agg([
        pl.col(var).mean().round(1).alias("mean"),
        pl.col(var).std().round(2).alias("sd"),
        pl.col(var).median().alias("median"),
        pl.col(var).min().alias("min"),
        pl.col(var).max().alias("max"),
        pl.len().alias("n")
    ])

```

```
age_stats = summarize_continuous(adsl, "AGE")
age_stats
```

| TRT01P                 | mean | sd   | median | min  | max  | n   |
|------------------------|------|------|--------|------|------|-----|
| str                    | f64  | f64  | f64    | f64  | f64  | u32 |
| "Xanomeline High Dose" | 74.4 | 7.89 | 76.0   | 56.0 | 88.0 | 84  |
| "Xanomeline Low Dose"  | 75.7 | 8.29 | 77.5   | 51.0 | 88.0 | 84  |
| "Placebo"              | 75.2 | 8.59 | 76.0   | 52.0 | 89.0 | 86  |

### 7.3.2 Categorical variables (sex, race)

For categorical variables, we calculate counts and percentages.

```
def summarize_categorical(df, var):
    """Calculate counts and percentages for categorical variables"""
    # Get counts by treatment and category
    counts = df.group_by(["TRT01P", var]).len()

    # Get treatment totals for percentage calculations
    totals = df.group_by("TRT01P").len().rename({"len": "total"})

    # Calculate percentages
    result = counts.join(totals, on="TRT01P").with_columns([
        (100.0 * pl.col("len") / pl.col("total")).round(1).alias("pct")
    ])

    return result

sex_stats = summarize_categorical(adsl, "SEX")
sex_stats
```

| TRT01P                 | SEX      | len | total | pct  |
|------------------------|----------|-----|-------|------|
| str                    | str      | u32 | u32   | f64  |
| "Xanomeline High Dose" | "Female" | 40  | 84    | 47.6 |
| "Placebo"              | "Male"   | 33  | 86    | 38.4 |
| "Xanomeline Low Dose"  | "Male"   | 34  | 84    | 40.5 |

| TRT01P<br>str          | SEX<br>str | len<br>u32 | total<br>u32 | pct<br>f64 |
|------------------------|------------|------------|--------------|------------|
| ...                    | ...        | ...        | ...          | ...        |
| "Xanomeline High Dose" | "Male"     | 44         | 84           | 52.4       |
| "Placebo"              | "Female"   | 53         | 86           | 61.6       |

```
race_stats = summarize_categorical(adsl, "RACE")
race_stats
```

| TRT01P<br>str          | RACE<br>str                         | len<br>u32 | total<br>u32 | pct<br>f64 |
|------------------------|-------------------------------------|------------|--------------|------------|
| "Xanomeline Low Dose"  | "Black Or African American"         | 6          | 84           | 7.1        |
| "Xanomeline High Dose" | "American Indian Or Alaska Nati..." | 1          | 84           | 1.2        |
| "Placebo"              | "White"                             | 78         | 86           | 90.7       |
| ...                    | ...                                 | ...        | ...          | ...        |
| "Xanomeline High Dose" | "Black Or African American"         | 9          | 84           | 10.7       |
| "Xanomeline Low Dose"  | "White"                             | 78         | 84           | 92.9       |

## 7.4 Step 3: Format results

Now we format the statistics into the standard baseline table format.

### 7.4.1 Format age statistics

```
# Format age as "Mean (SD)" and "Median [Min, Max]"
age_formatted = age_stats.with_columns([
    pl.format("{} ({})", pl.col("mean"), pl.col("sd")).alias("mean_sd"),
    pl.format("{} [{}, {}]", pl.col("median"), pl.col("min"), pl.col("max")).alias("median_range")
]).select(["TRT01P", "mean_sd", "median_range"])

age_formatted
```

| TRT01P                 | mean_sd       | median_range        |
|------------------------|---------------|---------------------|
| str                    | str           | str                 |
| "Xanomeline High Dose" | "74.4 (7.89)" | "76.0 [56.0, 88.0]" |
| "Xanomeline Low Dose"  | "75.7 (8.29)" | "77.5 [51.0, 88.0]" |
| "Placebo"              | "75.2 (8.59)" | "76.0 [52.0, 89.0]" |

## 7.4.2 Format categorical statistics

```
# Format categorical as "n (%)"
sex_formatted = sex_stats.with_columns(
    pl.format("{} ({}%)", pl.col("len"), pl.col("pct")).alias("n_pct")
).select(["TRT01P", "SEX", "n_pct"])

race_formatted = race_stats.with_columns(
    pl.format("{} ({}%)", pl.col("len"), pl.col("pct")).alias("n_pct")
).select(["TRT01P", "RACE", "n_pct"])

sex_formatted
```

| TRT01P                 | SEX      | n_pct        |
|------------------------|----------|--------------|
| str                    | str      | str          |
| "Xanomeline High Dose" | "Female" | "40 (47.6%)" |
| "Placebo"              | "Male"   | "33 (38.4%)" |
| "Xanomeline Low Dose"  | "Male"   | "34 (40.5%)" |
| ...                    | ...      | ...          |
| "Xanomeline High Dose" | "Male"   | "44 (52.4%)" |
| "Placebo"              | "Female" | "53 (61.6%)" |

## 7.5 Step 4: Create table structure

We'll build the table row by row following the standard baseline table format.

```

# Helper function to get value for a treatment group
def get_value(df, treatment):
    """Get value for a specific treatment group or return default"""
    result = df.filter(pl.col("TRT01P") == treatment)
    return result[result.columns[-1]][0] if result.height > 0 else "0 (0.0%)"

# Build the baseline table structure
table_rows = []

# Age section
table_rows.append(["Age (years)", "", "", ""])

# Age Mean (SD) row
age_mean_row = [" Mean (SD)"] + [
    get_value(age_formatted.select(["TRT01P", "mean_sd"]), trt).replace("0 (0.0%)", "")
    for trt in ["Placebo", "Xanomeline Low Dose", "Xanomeline High Dose"]
]
table_rows.append(age_mean_row)

# Age Median [Min, Max] row
age_median_row = [" Median [Min, Max]"] + [
    get_value(age_formatted.select(["TRT01P", "median_range"]), trt).replace("0 (0.0%)", "")
    for trt in ["Placebo", "Xanomeline Low Dose", "Xanomeline High Dose"]
]
table_rows.append(age_median_row)

# Sex section
table_rows.append(["Sex", "", "", ""])

for sex_cat in ["Female", "Male"]:
    sex_data = sex_formatted.filter(pl.col("SEX") == sex_cat)
    sex_row = [f" {sex_cat}"] + [
        get_value(sex_data, trt)
        for trt in ["Placebo", "Xanomeline Low Dose", "Xanomeline High Dose"]
    ]
    table_rows.append(sex_row)

# Race section
table_rows.append(["Race", "", "", ""])

```

```

for race_cat in ["White", "Black Or African American", "American Indian Or Alaska Native"]:
    race_data = race_formatted.filter(pl.col("RACE") == race_cat)
    race_row = [f" {race_cat}"] + [
        get_value(race_data, trt)
        for trt in ["Placebo", "Xanomeline Low Dose", "Xanomeline High Dose"]
    ]
    table_rows.append(race_row)

# Create DataFrame from table rows
baseline_table = pl.DataFrame(
    table_rows,
    schema=["Characteristic", "Placebo", "Xanomeline Low Dose", "Xanomeline High Dose"],
    orient="row"
)

baseline_table

```

| Characteristic<br>str             | Placebo<br>str      | Xanomeline Low Dose<br>str | Xanomeline High Dose<br>str |
|-----------------------------------|---------------------|----------------------------|-----------------------------|
| "Age (years)"                     | ""                  | ""                         | ""                          |
| " Mean (SD)"                      | "75.2 (8.59)"       | "75.7 (8.29)"              | "74.4 (7.89)"               |
| " Median [Min, Max]"              | "76.0 [52.0, 89.0]" | "77.5 [51.0, 88.0]"        | "76.0 [56.0, 88.0]"         |
| ...                               | ...                 | ...                        | ...                         |
| " Black Or African American"      | "8 (9.3%)"          | "6 (7.1%)"                 | "9 (10.7%)"                 |
| " American Indian Or Alaska Na... | "0 (0.0%)"          | "0 (0.0%)"                 | "1 (1.2%)"                  |

## 7.6 Step 5: Generate publication-ready output

Finally, we format the baseline table for regulatory submission using the `rtflite` package.

```

# Get treatment group sizes for column headers
treatment_n = adsl.group_by("TRT01P").len().sort("TRT01P")
n_placebo = treatment_n.filter(pl.col("TRT01P") == "Placebo")["len"][0]
n_low = treatment_n.filter(pl.col("TRT01P") == "Xanomeline Low Dose")["len"][0]
n_high = treatment_n.filter(pl.col("TRT01P") == "Xanomeline High Dose")["len"][0]

```

```

doc_baseline = rtf.RTFDocument(
    df=baseline_table,
    rtf_title=rtf.RTFTitle(
        text=[
            "Baseline Characteristics of Participants",
            "(All Participants Randomized)"
        ]
    ),
    rtf_column_header=rtf.RTFColumnHeader(
        text=[
            "Characteristic",
            f"Placebo\n(N={n_placebo})",
            f"Xanomeline Low Dose\n(N={n_low})",
            f"Xanomeline High Dose\n(N={n_high})"
        ],
        text_justification=["l", "c", "c", "c"],
        col_rel_width=[3, 2, 2, 2]
    ),
    rtf_body=rtf.RTFBody(
        text_justification=["l", "c", "c", "c"],
        col_rel_width=[3, 2, 2, 2]
    ),
    rtf_source=rtf.RTFSource(text=["Source: ADSL dataset"])
)

doc_baseline.write_rtf("rtf/tlf_baseline.rtf") # Save as RTF for submission

rtf/tlf_baseline.rtf

PosixPath('pdf/tlf_baseline.pdf')

```

## 8 Adverse events summary

### Objective

Create adverse event summary tables to provide high-level safety overview across treatment groups. Learn to calculate AE rates and percentages using Polars and create comprehensive safety summary tables with `rtflite`.

### 8.1 Overview

Adverse events (AE) summary tables are critical safety assessments required in clinical study reports. Following [ICH E3 guidance](#), these tables summarize the overall safety profile by showing the number and percentage of participants experiencing various categories of adverse events across treatment groups.

Key categories typically include:

- **Any adverse event:** Total participants with at least one AE
- **Drug-related events:** Events potentially related to study treatment
- **Serious adverse events:** Events meeting regulatory criteria for seriousness
- **Deaths:** Fatal outcomes
- **Discontinuations:** Participants who stopped treatment due to AEs

This tutorial shows you how to create an AE summary table using Python's `rtflite` package.



```
import polars as pl
import rtflite as rtf
```

```
polars.config.Config
```

## 8.2 Step 1: Load data

We need two datasets for AE analysis: the subject-level dataset (ADSL) and the adverse events dataset (ADAE).

```
# Load datasets
adsl = pl.read_parquet("data/adsl.parquet")
adae = pl.read_parquet("data/adae.parquet")

# Display key variables from ADSL
adsl.select(["USUBJID", "TRT01A", "SAFFL"])
```

| USUBJID       | TRT01A                 | SAFFL |
|---------------|------------------------|-------|
| str           | str                    | str   |
| "01-701-1015" | "Placebo"              | "Y"   |
| "01-701-1023" | "Placebo"              | "Y"   |
| "01-701-1028" | "Xanomeline High Dose" | "Y"   |
| ...           | ...                    | ...   |
| "01-718-1371" | "Xanomeline High Dose" | "Y"   |
| "01-718-1427" | "Xanomeline High Dose" | "Y"   |

```
# Display key variables from ADAE
adae.select(["USUBJID", "AEREL", "AESER", "AEOUT", "AEACN"])
```

| USUBJID       | AEREL      | AESER | AEOUT                        | AEACN |
|---------------|------------|-------|------------------------------|-------|
| str           | str        | str   | str                          | str   |
| "01-701-1015" | "PROBABLE" | "N"   | "NOT RECOVERED/NOT RESOLVED" | " "   |
| "01-701-1015" | "PROBABLE" | "N"   | "NOT RECOVERED/NOT RESOLVED" | " "   |
| "01-701-1015" | "REMOTE"   | "N"   | "RECOVERED/RESOLVED"         | " "   |
| ...           | ...        | ...   | ...                          | ...   |

| USUBJID       | AEREL      | AESER | AEOUT                | AEACN |
|---------------|------------|-------|----------------------|-------|
| str           | str        | str   | str                  | str   |
| "01-718-1427" | "POSSIBLE" | "N"   | "RECOVERED/RESOLVED" | " "   |
| "01-718-1427" | "POSSIBLE" | "N"   | "RECOVERED/RESOLVED" | " "   |

Key ADAE variables used in this analysis:

- **USUBJID**: Unique subject identifier to link with ADSL
- **AEREL**: Relationship of adverse event to study drug (e.g., "RELATED", "POSSIBLE", "PROBABLE", "DEFINITE", "NOT RELATED")
- **AESER**: Serious adverse event flag ("Y" = serious, "N" = not serious)
- **AEOUT**: Outcome of adverse event (e.g., "RECOVERED", "RECOVERING", "NOT RECOVERED", "FATAL")
- **AEACN**: Action taken with study treatment (e.g., "DOSE NOT CHANGED", "DRUG WITHDRAWN", "DOSE REDUCED")

## 8.3 Step 2: Filter safety population

For safety analyses, we focus on participants who received at least one dose of study treatment.

```
# Filter to safety population
adsl_safety = adsl.filter(pl.col("SAFFL") == "Y").select(["USUBJID", "TRT01A"])

# Get treatment counts for denominators
pop_counts = adsl_safety.group_by("TRT01A").agg(
    N = pl.len()
).sort("TRT01A")

# Preserve the treatment level order for downstream joins
treatment_levels = pop_counts.select(["TRT01A"])

# Safety population by treatment
pop_counts
```

|                        |     |
|------------------------|-----|
| TRT01A                 | N   |
| str                    | u32 |
| "Placebo"              | 86  |
| "Xanomeline High Dose" | 84  |
| "Xanomeline Low Dose"  | 84  |

```
# Join treatment information to AE data
adae_safety = adae.join(adsl_safety, on="USUBJID")

# Total AE records in safety population
adae_safety.height
```

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## 8.4 Step 3: Define AE categories

We'll calculate participant counts for standard AE categories used in regulatory submissions.

```
def count_participants(df, condition=None):
    """
    Count unique participants meeting a condition

    Args:
        df: DataFrame with adverse events
        condition: polars expression for filtering (None = count all)

    Returns:
        DataFrame with counts by treatment
    """
    if condition is not None:
        df = df.filter(condition)

    counts = df.group_by("TRT01A").agg(
        n = pl.col("USUBJID").n_unique()
    )
```

```

    return treatment_levels.join(counts, on="TRT01A", how="left").with_columns(
        pl.col("n").fill_null(0)
    )

# Calculate each category
categories = []

# 1. Participants in population (no filtering)
pop_row = pop_counts.with_columns(
    category = pl.lit("Participants in population")
).rename({"N": "n"})
categories.append(pop_row)

# 2. With any adverse event
any_ae = count_participants(adae_safety).with_columns(
    category = pl.lit("With any adverse event")
)
categories.append(any_ae)

# 3. With drug-related adverse event
drug_related = count_participants(
    adae_safety,
    pl.col("AEREL").is_in(["POSSIBLE", "PROBABLE", "DEFINITE", "RELATED"])
).with_columns(
    category = pl.lit("With drug-related adverse event")
)
categories.append(drug_related)

# 4. With serious adverse event
serious = count_participants(
    adae_safety,
    pl.col("AESER") == "Y"
).with_columns(
    category = pl.lit("With serious adverse event")
)
categories.append(serious)

# 5. With serious drug-related adverse event
serious_drug_related = count_participants(
    adae_safety,

```

```

    (pl.col("AESER") == "Y") &
    pl.col("AEREL").is_in(["POSSIBLE", "PROBABLE", "DEFINITE", "RELATED"])
  ).with_columns(
    category = pl.lit("With serious drug-related adverse event")
  )
categories.append(serious_drug_related)

# 6. Who died
deaths = count_participants(
  adae_safety,
  pl.col("AEOUT") == "FATAL"
).with_columns(
  category = pl.lit("Who died")
)
categories.append(deaths)

# 7. Discontinued due to adverse event
discontinued = count_participants(
  adae_safety,
  pl.col("AEACN") == "DRUG WITHDRAWN"
).with_columns(
  category = pl.lit("Discontinued due to adverse event")
)
categories.append(discontinued)

```

## 8.5 Step 4: Combine and calculate percentages

Now we combine all categories and calculate percentages based on the safety population.

```

# Combine all categories
ae_summary = pl.concat(categories, how="diagonal")

# Add population totals and calculate percentages
ae_summary = ae_summary.join(
  pop_counts.select(["TRT01A", "N"]),
  on="TRT01A",

```

```

    how="left"
  ).with_columns([
    # Fill missing counts with 0
    pl.col("n").fill_null(0),
    # Calculate percentage
    pl.when(pl.col("category") == "Participants in population")
      .then(None) # No percentage for population row
      .otherwise((100.0 * pl.col("n") / pl.col("N")).round(1))
      .alias("pct")
  ])

ae_summary.sort(["category", "TRT01A"])

```

| TRT01A                 | n   | category                           | N   | pct |
|------------------------|-----|------------------------------------|-----|-----|
| str                    | u32 | str                                | u32 | f64 |
| "Placebo"              | 0   | "Discontinued due to adverse ev... | 86  | 0.0 |
| "Xanomeline High Dose" | 0   | "Discontinued due to adverse ev... | 84  | 0.0 |
| "Xanomeline Low Dose"  | 0   | "Discontinued due to adverse ev... | 84  | 0.0 |
| ...                    | ... | ...                                | ... | ... |
| "Xanomeline High Dose" | 1   | "With serious drug-related adve... | 84  | 1.2 |
| "Xanomeline Low Dose"  | 1   | "With serious drug-related adve... | 84  | 1.2 |

## 8.6 Step 5: Format for display

We'll format the counts and percentages for the final table display.

```

# Format display values
ae_formatted = ae_summary.with_columns([
    # Show counts as strings, including zeros
    pl.col("n").cast(str).alias("n_display"),
    # Format percentages with parentheses; blank out population row
    pl.when(pl.col("category") == "Participants in population")
      .then(pl.lit(""))
      .otherwise(
        pl.format("({})", pl.col("pct").fill_null(0).round(1).cast(str))
      )
])

```

```

        .alias("pct_display")
    ])

ae_formatted.select(["category", "TRT01A", "n_display", "pct_display"])

```

| category                           | TRT01A                 | n_display | pct_display |
|------------------------------------|------------------------|-----------|-------------|
| str                                | str                    | str       | str         |
| "Participants in population"       | "Placebo"              | "86"      | ""          |
| "Participants in population"       | "Xanomeline High Dose" | "84"      | ""          |
| "Participants in population"       | "Xanomeline Low Dose"  | "84"      | ""          |
| ...                                | ...                    | ...       | ...         |
| "Discontinued due to adverse ev... | "Xanomeline High Dose" | "0"       | "(0.0)"     |
| "Discontinued due to adverse ev... | "Xanomeline Low Dose"  | "0"       | "(0.0)"     |

## 8.7 Step 6: Create final table structure

We reshape the data to create the final table with treatments as columns.

```

# Define category order for consistent display
category_order = [
    "Participants in population",
    "With any adverse event",
    "With drug-related adverse event",
    "With serious adverse event",
    "With serious drug-related adverse event",
    "Who died",
    "Discontinued due to adverse event"
]

# Pivot to wide format
ae_wide = ae_formatted.pivot(
    values=["n_display", "pct_display"],
    index="category",
    on="TRT01A",
    maintain_order=True
)

```

```
# Reorder columns for each treatment group
treatments = ["Placebo", "Xanomeline Low Dose", "Xanomeline High Dose"]
column_order = ["category"]
for trt in treatments:
    column_order.extend([f"n_display_{trt}", f"pct_display_{trt}"])

# Create final table with proper column order
final_table = ae_wide.select(column_order).sort(
    pl.col("category").cast(pl.Enum(category_order))
)

final_table
```

| category<br>str                     | n_display_Placebo<br>str | pct_display_Placebo<br>str | n_display_Xanomeline Low D<br>str |
|-------------------------------------|--------------------------|----------------------------|-----------------------------------|
| "Participants in population"        | "86"                     | "                          | "84"                              |
| "With any adverse event"            | "69"                     | "(80.2)"                   | "77"                              |
| "With drug-related adverse even..." | "44"                     | "(51.2)"                   | "73"                              |
| ...                                 | ...                      | ...                        | ...                               |
| "Who died"                          | "2"                      | "(2.3)"                    | "1"                               |
| "Discontinued due to adverse ev..." | "0"                      | "(0.0)"                    | "0"                               |

## 8.8 Step 7: Generate publication-ready output

Finally, we format the AE summary table for regulatory submission using the `rtflite` package.

```
# Get population sizes for column headers
n_placebo = pop_counts.filter(pl.col("TRT01A") == "Placebo")["N"][0]
n_low = pop_counts.filter(pl.col("TRT01A") == "Xanomeline Low Dose")["N"][0]
n_high = pop_counts.filter(pl.col("TRT01A") == "Xanomeline High Dose")["N"][0]

doc_ae_summary = rtf.RTFDocument(
    df=final_table.rename({"category": ""}),
    rtf_title=rtf.RTFTitle(
```



```

        text=[
            "Analysis of Adverse Event Summary",
            "(Safety Analysis Population)"
        ]
    ),
    rtf_column_header=[
        rtf.RTFColumnHeader(
            text = [
                "",
                "Placebo",
                "Xanomeline Low Dose",
                "Xanomeline High Dose"
            ],
            col_rel_width=[4, 2, 2, 2],
            text_justification=["l", "c", "c", "c"],
        ),
        rtf.RTFColumnHeader(
            text=[
                "",          # Empty for first column
                "n", "(%)",  # Placebo columns
                "n", "(%)",  # Low Dose columns
                "n", "(%)"   # High Dose columns
            ],
            col_rel_width=[4] + [1] * 6,
            text_justification=["l"] + ["c"] * 6,
            border_left = ["single"] + ["single", ""] * 3,
            border_top = [""] + ["single"] * 6
        )
    ],
    rtf_body=rtf.RTFBody(
        col_rel_width=[4] + [1] * 6,
        text_justification=["l"] + ["c"] * 6,
        border_left = ["single"] + ["single", ""] * 3
    ),
    rtf_footnote=rtf.RTFFootnote(
        text=[
            "Every subject is counted a single time for each applicable row and column."
        ]
    ),
    rtf_source=rtf.RTFSource(

```

```
        text=["Source: ADSL and ADAE datasets"]
    )
)

doc_ae_summary.write_rtf("rtf/tlf_ae_summary.rtf")
```

rtf/tlf\_ae\_summary.rtf

PosixPath('pdf/tlf\_ae\_summary.pdf')

## 9 Specific adverse events

### Objective

Create detailed adverse event tables organized by System Organ Class and Preferred Term to support safety evaluation. Learn to process ADAE data with hierarchical grouping using Polars and generate regulatory-compliant AE listings with `rtflite`.

### 9.1 Overview

Specific adverse events tables provide detailed safety information organized by System Organ Class (SOC) and Preferred Term (PT) following the Medical Dictionary for Regulatory Activities (MedDRA) hierarchy. Following [ICH E3 guidance](#), these tables are essential components of clinical study reports that present participant-level adverse event data across treatment groups.

Key features of specific AE tables include:

- **Hierarchical structure:** SOC categories with nested specific AE terms
- **Participant counts:** Number of participants experiencing each AE type
- **Treatment comparison:** Side-by-side counts across treatment groups
- **MedDRA compliance:** Standardized medical terminology for regulatory submissions

This tutorial demonstrates how to create a regulatory-compliant specific adverse events table using Python's `rtflite` package.

## 9.2 Setup

```
import polars as pl
import rtflite as rtf
```

```
polars.config.Config
```

```
adsl = pl.read_parquet("data/adsl.parquet")
adae = pl.read_parquet("data/adae.parquet")
treatments = ["Placebo", "Xanomeline Low Dose", "Xanomeline High Dose"]
```

## 9.3 Step 1: Load and explore data

We start by examining the adverse events data structure and understanding the MedDRA hierarchy.

```
# Display key variables in ADAE dataset
adae_vars = adae.select(["USUBJID", "TRTA", "AEBODSYS", "AEDECOD", "AESEV", "AESER"])
# Key ADAE variables
adae_vars
```

| USUBJID       | TRTA                   | AEBODSYS                           | AEDECOD         |
|---------------|------------------------|------------------------------------|-----------------|
| str           | str                    | str                                | str             |
| "01-701-1015" | "Placebo"              | "GENERAL DISORDERS AND ADMINIST... | "APPLICATION SI |
| "01-701-1015" | "Placebo"              | "GENERAL DISORDERS AND ADMINIST... | "APPLICATION SI |
| "01-701-1015" | "Placebo"              | "GASTROINTESTINAL DISORDERS"       | "DIARRHOEA"     |
| ...           | ...                    | ...                                | ...             |
| "01-718-1427" | "Xanomeline High Dose" | "METABOLISM AND NUTRITION DISOR... | "DECREASED AP   |
| "01-718-1427" | "Xanomeline High Dose" | "GASTROINTESTINAL DISORDERS"       | "NAUSEA"        |

```
# Examine the MedDRA hierarchy structure
# System Organ Classes (SOCs) in the data
soc_summary = adae.group_by("AEBODSYS").agg(
    n_participants=pl.col("USUBJID").n_unique(),
    n_events=pl.len()
```

```
).sort("n_participants", descending=True)
soc_summary
```

| AEBODSYS<br>str                    | n_participants<br>u32 | n_events<br>u32 |
|------------------------------------|-----------------------|-----------------|
| "GENERAL DISORDERS AND ADMINIST... | 108                   | 292             |
| "SKIN AND SUBCUTANEOUS TISSUE D... | 105                   | 276             |
| "NERVOUS SYSTEM DISORDERS"         | 59                    | 101             |
| ...                                | ...                   | ...             |
| "HEPATOBIILIARY DISORDERS"         | 1                     | 1               |
| "SOCIAL CIRCUMSTANCES"             | 1                     | 1               |

## 9.4 Step 2: Prepare analysis population

Following regulatory standards, we focus on the safety analysis population.

```
# Define safety population
adsl_safety = adsl.filter(pl.col("SAFFL") == "Y").select(["USUBJID", "TRT01A"])
# Safety population size
adsl_safety.height

# Get safety population counts by treatment
pop_counts = adsl_safety.group_by("TRT01A").agg(N=pl.len()).sort("TRT01A")
# Safety population by treatment
pop_counts
```

| TRT01A<br>str          | N<br>u32 |
|------------------------|----------|
| "Placebo"              | 86       |
| "Xanomeline High Dose" | 84       |
| "Xanomeline Low Dose"  | 84       |

```
# Filter adverse events to safety population
adae_safety = adae.join(adsl_safety, on="USUBJID", how="inner")
# AE records in safety population
adae_safety.height
```

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## 9.5 Step 3: Data preparation and standardization

We standardize the adverse event terms and prepare the hierarchical data structure.

```
# Standardize AE term formatting for consistency
ae_counts = (
    adae_safety
    .with_columns([
        pl.col("AEDECOD").str.to_titlecase().alias("AEDECOD_STD"),
        pl.col("AEBODSYS").str.to_titlecase().alias("AEBODSYS_STD")
    ])
    .group_by(["TRT01A", "AEBODSYS_STD", "AEDECOD_STD"])
    .agg(n=pl.col("USUBJID").n_unique())
    .sort(["AEBODSYS_STD", "AEDECOD_STD", "TRT01A"])
)

# Sample of prepared AE counts
ae_counts
```

| TRT01A                 | AEBODSYS_STD         | AEDECOD_STD               | n   |
|------------------------|----------------------|---------------------------|-----|
| str                    | str                  | str                       | u32 |
| "Placebo"              | "Cardiac Disorders"  | "Atrial Fibrillation"     | 1   |
| "Xanomeline High Dose" | "Cardiac Disorders"  | "Atrial Fibrillation"     | 3   |
| "Xanomeline Low Dose"  | "Cardiac Disorders"  | "Atrial Fibrillation"     | 1   |
| ...                    | ...                  | ...                       | ... |
| "Placebo"              | "Vascular Disorders" | "Orthostatic Hypotension" | 1   |
| "Xanomeline High Dose" | "Vascular Disorders" | "Wound Haemorrhage"       | 1   |

## 9.6 Step 4: Build hierarchical table structure

We create a nested table structure with SOC headers and indented specific terms.

```
# Initialize table with population counts
table_data = [
    ["Participants in population"] + [
        str(pop_counts.filter(pl.col("TRT01A") == t)["N"][0])
        for t in treatments
    ],
    [""] * 4 # Blank separator row
]

# Build hierarchical structure: SOC -> Specific AE terms
for soc in ae_counts["AEBODSYS_STD"].unique().sort():
    # Add SOC header row (bold formatting will be applied later)
    table_data.append([soc] + [""] * 3)

    # Get all AE terms within this SOC
    soc_data = ae_counts.filter(pl.col("AEBODSYS_STD") == soc)

    # Add each specific AE term with counts
    for ae_term in soc_data["AEDECOD_STD"].unique().sort():
        row = [f" {ae_term}"] # Indent specific terms

        # Add counts for each treatment group
        for trt in treatments:
            count_data = soc_data.filter(
                (pl.col("AEDECOD_STD") == ae_term) &
                (pl.col("TRT01A") == trt)
            )
            count = count_data["n"][0] if count_data.height > 0 else 0
            row.append(str(count))

        table_data.append(row)

# Convert to Polars DataFrame
df_ae_specific = pl.DataFrame(
    table_data,
```

```

    schema=[""] + treatments,
    orient="row"
)

# Final table structure
df_ae_specific.shape
df_ae_specific

```

| column_0                     | Placebo | Xanomeline Low Dose | Xanomeline High Dose |
|------------------------------|---------|---------------------|----------------------|
| str                          | str     | str                 | str                  |
| "Participants in population" | "86"    | "84"                | "84"                 |
| "                            | "       | "                   | "                    |
| "Cardiac Disorders"          | "       | "                   | "                    |
| ...                          | ...     | ...                 | ...                  |
| " Orthostatic Hypotension"   | "1"     | "0"                 | "0"                  |
| " Wound Haemorrhage"         | "0"     | "0"                 | "1"                  |

## 9.7 Step 5: Create regulatory-compliant RTF output

We format the table following regulatory submission standards with proper hierarchy and formatting.

```

# Create comprehensive RTF document
doc_ae_specific = rtf.RTFDocument(
    df=df_ae_specific,
    rtf_title=rtf.RTFTitle(
        text=[
            "Adverse Events by System Organ Class and Preferred Term",
            "(Safety Analysis Set)"
        ]
    ),
    rtf_column_header=rtf.RTFColumnHeader(
        text=[
            "System Organ Class\\line Preferred Term",
            f"Placebo\\line (N={pop_counts.filter(pl.col('TRT01A') == 'Placebo')['N'][0]}))",
            f"Xanomeline Low Dose\\line (N={pop_counts.filter(pl.col('TRT01A') == 'Xanomeline L

```



```

        f"Xanomeline High Dose\\line (N={pop_counts.filter(pl.col('TRT01A') == 'Xanomeline
    ],
    col_rel_width=[4, 1.5, 1.5, 1.5],
    text_justification=["l", "c", "c", "c"],
    text_format="b", # Bold headers
    border_bottom="single"
),
rtf_body=rtf.RTFBody(
    col_rel_width=[4, 1.5, 1.5, 1.5],
    text_justification=["l", "c", "c", "c"],
    # Apply bold formatting to SOC headers (rows without indentation)
    text_font_style=lambda df, i, j: "b" if j == 0 and not str(df[i, j]).startswith(" ")
),
rtf_footnote=rtf.RTFFootnote(
    text=[
        "MedDRA version 25.0.",
        "Each participant is counted once within each preferred term and system organ class",
        "Participants with multiple events in the same preferred term are counted only once"
    ]
),
rtf_source=rtf.RTFSource(
    text=["Source: ADAE Analysis Dataset (Data cutoff: 01JAN2023)"]
)
)

# Generate RTF file
doc_ae_specific.write_rtf("rtf/tlf_ae_specific.rtf")
# RTF file created: rtf/tlf_ae_specific.rtf

```

rtf/tlf\_ae\_specific.rtf

PosixPath('pdf/tlf\_ae\_specific.pdf')

# 10 ANCOVA efficacy analysis

## Objective

Create ANCOVA efficacy analysis tables to evaluate treatment effects while controlling for baseline covariates. Learn to perform ANCOVA modeling with missing data imputation using Python statistical libraries and present results in regulatory format with rtfite.

## 10.1 Overview

Analysis of Covariance (ANCOVA) is the primary statistical method for efficacy evaluation in clinical trials. Following [ICH E9 guidance](#) on statistical principles for clinical trials, ANCOVA provides a robust framework for comparing treatment effects while controlling for baseline covariates.

Key features of ANCOVA efficacy analysis include:

- **Covariate adjustment:** Controls for baseline values to reduce variability
- **Least squares means:** Provides treatment effect estimates adjusted for covariates
- **Missing data handling:** Implements appropriate imputation strategies (e.g., LOCF, MMRM)
- **Pairwise comparisons:** Tests specific treatment contrasts with confidence intervals
- **Regulatory compliance:** Follows statistical analysis plan specifications

This tutorial demonstrates how to create a comprehensive ANCOVA efficacy table for glucose change from baseline us-

ing Python’s statistical libraries and `rtflite` for regulatory-compliant formatting.

## 10.2 Setup

```
import polars as pl
import rtflite as rtf
import pandas as pd
import numpy as np
import statsmodels.formula.api as smf
from scipy import stats as scipy_stats
```

```
polars.config.Config
```

```
adsl = pl.read_parquet("data/adsl.parquet")
adlbc = pl.read_parquet("data/adlbc.parquet")
treatments = ["Placebo", "Xanomeline Low Dose", "Xanomeline High Dose"]
```

## 10.3 Step 1: Explore laboratory data structure

We start by understanding the glucose data structure and endpoint definitions.

```
# Display key laboratory variables
# Key ADLBC variables for efficacy analysis
lab_vars = adlbc.select(["USUBJID", "PARAMCD", "PARAM", "AVISIT", "AVISITN", "AVAL", "BASE", "C"]
lab_vars
```

| USUBJID       | PARAMCD  | PARAM                | AVISIT |           | AVISITN | AVAL  | BASE  |
|---------------|----------|----------------------|--------|-----------|---------|-------|-------|
| str           | str      | str                  | str    |           | f64     | f64   | f64   |
| "01-701-1015" | "SODIUM" | "Sodium (mmol/L)"    | "      | Baseline" | 0.0     | 140.0 | 140.0 |
| "01-701-1015" | "K"      | "Potassium (mmol/L)" | "      | Baseline" | 0.0     | 4.5   | 4.5   |
| "01-701-1015" | "CL"     | "Chloride (mmol/L)"  | "      | Baseline" | 0.0     | 106.0 | 106.0 |

| USUBJID       | PARAMCD | PARAM                              | AVISIT             | AVISITN | AVAL | B   |
|---------------|---------|------------------------------------|--------------------|---------|------|-----|
| str           | str     | str                                | str                | f64     | f64  | f64 |
| ...           | ...     | ...                                | ...                | ...     | ...  | ... |
| "01-718-1427" | "_CK"   | "Creatine Kinase (U/L) change f... | "Week 8"           | 8.0     | -0.5 | nu  |
| "01-718-1427" | "_CK"   | "Creatine Kinase (U/L) change f... | "End of Treatment" | 99.0    | -0.5 | nu  |

```
# Focus on glucose parameter
gluc_visits = adlbc.filter(pl.col("PARAMCD") == "GLUC").select("AVISIT", "AVISITN").unique().sort("AVISIT")
# Available glucose measurement visits
gluc_visits
```

| AVISIT             | AVISITN |
|--------------------|---------|
| str                | f64     |
| ...                | ...     |
| "Baseline"         | 0.0     |
| "Week 2"           | 2.0     |
| ...                | ...     |
| "Week 26"          | 26.0    |
| "End of Treatment" | 99.0    |

## 10.4 Step 2: Define analysis population and endpoint

Following the Statistical Analysis Plan, we focus on the efficacy population for the primary endpoint.

```
# Clean data types and prepare datasets
adlbc_clean = adlbc.with_columns([
    pl.col(c).cast(str).str.strip_chars()
    for c in ["USUBJID", "PARAMCD", "AVISIT", "TRTP"]
])

# Define efficacy population
adsl_eff = adsl.filter(pl.col("EFFFL") == "Y").select(["USUBJID"])
# Efficacy population size
adsl_eff.height
```

```
# Filter laboratory data to efficacy population
adlbc_eff = adlbc_clean.join(adsl_eff, on="USUBJID", how="inner")
# Laboratory records in efficacy population
adlbc_eff.height
```

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```
# Examine glucose data availability by visit and treatment
gluc_availability = (
  adlbc_eff.filter(pl.col("PARAMCD") == "GLUC")
    .group_by(["TRTP", "AVISIT"])
    .agg(n_subjects=pl.col("USUBJID").n_unique())
    .sort(["TRTP", "AVISIT"])
)
# Glucose data availability by visit
gluc_availability
```

| TRTP                  | AVISIT             | n_subjects |
|-----------------------|--------------------|------------|
| str                   | str                | u32        |
| "Placebo"             | "                  | 13         |
| "Placebo"             | "Baseline"         | 79         |
| "Placebo"             | "End of Treatment" | 79         |
| ...                   | ...                | ...        |
| "Xanomeline Low Dose" | "Week 6"           | 62         |
| "Xanomeline Low Dose" | "Week 8"           | 59         |

## 10.5 Step 3: Implement LOCF imputation strategy

We apply Last Observation Carried Forward (LOCF) for missing Week 24 glucose values.

```
# Prepare glucose data with LOCF for Week 24 endpoint
gluc_data = (
  adlbc_eff
    .filter((pl.col("PARAMCD") == "GLUC") & (pl.col("AVISITN") <= 24))
```

```

.sort(["USUBJID", "AVISITN"])
.group_by("USUBJID")
.agg([
    pl.col("TRTP").first(),
    pl.col("BASE").first(),
    pl.col("AVAL").filter(pl.col("AVISITN") == 0).first().alias("Baseline"),
    pl.col("AVAL").last().alias("Week_24_LOCF"), # LOCF: last available value <= Week 24
    pl.col("AVISITN").max().alias("Last_Visit") # Track actual last visit
])
.filter(pl.col("Baseline").is_not_null() & pl.col("Week_24_LOCF").is_not_null())
.with_columns((pl.col("Week_24_LOCF") - pl.col("Baseline")).alias("CHG"))
)

# Subjects with baseline and Week 24 (LOCF) glucose
gluc_data.height
# Sample of prepared analysis data
gluc_data

```

| USUBJID       | TRTP                   | BASE    | Baseline | Week_24_LOCF | Last_Visit | CHG      |
|---------------|------------------------|---------|----------|--------------|------------|----------|
| str           | str                    | f64     | f64      | f64          | f64        | f64      |
| "01-701-1015" | "Placebo"              | 4.71835 | 4.71835  | 4.49631      | 24.0       | -0.22204 |
| "01-701-1023" | "Placebo"              | 5.32896 | 5.32896  | 5.43998      | 4.0        | 0.11102  |
| "01-701-1028" | "Xanomeline High Dose" | 4.77386 | 4.77386  | 5.43998      | 24.0       | 0.66612  |
| ...           | ...                    | ...     | ...      | ...          | ...        | ...      |
| "01-718-1371" | "Xanomeline High Dose" | 6.27263 | 6.27263  | 5.10692      | 12.0       | -1.16571 |
| "01-718-1427" | "Xanomeline High Dose" | 4.55182 | 4.55182  | 3.71917      | 8.0        | -0.83265 |

```

# Assess LOCF imputation impact
locf_summary = (
    gluc_data
    .group_by(["TRTP", "Last_Visit"])
    .agg(n_subjects=pl.len())
    .sort(["TRTP", "Last_Visit"])
)
# LOCF imputation summary (last actual visit used)
locf_summary

```

| TRTP<br>str           | Last_Visit<br>f64 | n_subjects<br>u32 |
|-----------------------|-------------------|-------------------|
| "Placebo"             | 2.0               | 2                 |
| "Placebo"             | 4.0               | 2                 |
| "Placebo"             | 6.0               | 2                 |
| ...                   | ...               | ...               |
| "Xanomeline Low Dose" | 20.0              | 5                 |
| "Xanomeline Low Dose" | 24.0              | 25                |

## 10.6 Step 4: Calculate descriptive statistics

We compute baseline, Week 24, and change from baseline statistics by treatment group.

```
# Calculate comprehensive descriptive statistics
desc_stats = []
for trt in treatments:
    # Analysis data for this treatment
    trt_data = gluc_data.filter(pl.col("TRTP") == trt)

    # Original baseline data (all subjects with baseline)
    baseline_full = adlbc_eff.filter(
        (pl.col("PARAMCD") == "GLUC") &
        (pl.col("AVISIT") == "Baseline") &
        (pl.col("TRTP") == trt)
    )

    desc_stats.append({
        "Treatment": trt,
        "N_Baseline": baseline_full.height,
        "Baseline_Mean": baseline_full["AVAL"].mean() if baseline_full.height > 0 else np.nan,
        "Baseline_SD": baseline_full["AVAL"].std() if baseline_full.height > 0 else np.nan,
        "N_Week24": trt_data.height,
        "Week24_Mean": trt_data["Week_24_LOCF"].mean() if trt_data.height > 0 else np.nan,
        "Week24_SD": trt_data["Week_24_LOCF"].std() if trt_data.height > 0 else np.nan,
        "N_Change": trt_data.height,
        "Change_Mean": trt_data["CHG"].mean() if trt_data.height > 0 else np.nan,
        "Change_SD": trt_data["CHG"].std() if trt_data.height > 0 else np.nan
    })
```

```

})

# Display descriptive statistics
desc_df = pl.DataFrame(desc_stats)
# Descriptive statistics by treatment
desc_df

```

| Treatment              | N_Baseline | Baseline_Mean | Baseline_SD | N_Week24 | Week24_Mean | Week24_SD |
|------------------------|------------|---------------|-------------|----------|-------------|-----------|
| str                    | i64        | f64           | f64         | i64      | f64         | f64       |
| "Placebo"              | 79         | 5.656399      | 2.229324    | 79       | 5.639535    | 1.651800  |
| "Xanomeline Low Dose"  | 79         | 5.419603      | 0.946102    | 79       | 5.352148    | 1.058000  |
| "Xanomeline High Dose" | 74         | 5.388971      | 1.374893    | 74       | 5.831551    | 2.214600  |

## 10.7 Step 5: Perform ANCOVA analysis

We fit the ANCOVA model with treatment and baseline glucose as covariates.

```

# Convert to pandas for statsmodels compatibility
ancova_df = gluc_data.to_pandas()
ancova_df["TRTP"] = pd.Categorical(ancova_df["TRTP"], categories=treatments)

# Fit ANCOVA model: Change = Treatment + Baseline
model = smf.ols("CHG ~ TRTP + BASE", data=ancova_df).fit()

# Display model summary
# ANCOVA Model Summary
model.rsquared
model.fvalue, model.f_pvalue
# Model coefficients
model.summary().tables[1]

```



|                                     | coef    | std err | t      | P>  t | [0.025 | 0.975] |
|-------------------------------------|---------|---------|--------|-------|--------|--------|
| <b>Intercept</b>                    | 2.9972  | 0.392   | 7.642  | 0.000 | 2.224  | 3.770  |
| <b>TRTP[T.Xanomeline Low Dose]</b>  | -0.1768 | 0.243   | -0.729 | 0.467 | -0.655 | 0.301  |
| <b>TRTP[T.Xanomeline High Dose]</b> | 0.3169  | 0.247   | 1.284  | 0.200 | -0.169 | 0.803  |
| <b>BASE</b>                         | -0.5329 | 0.062   | -8.543 | 0.000 | -0.656 | -0.410 |

```
# Calculate adjusted means (LS means) at mean baseline value
base_mean = ancova_df["BASE"].mean()
var_cov = model.cov_params()
ls_means = []

# LS means calculated at baseline mean
base_mean

for i, trt in enumerate(treatments):
    # Create prediction vector for LS mean calculation
    # Model: CHG = intercept + trt_effect1*(trt==1) + trt_effect2*(trt==2) + base_effect*baseline
    x_pred = np.array([1, int(i==1), int(i==2), base_mean])

    # Calculate LS mean
    ls_mean = model.predict(pd.DataFrame({"TRTP": [trt], "BASE": [base_mean]}))[0]

    # Calculate standard error for confidence interval
    se_pred = np.sqrt(x_pred @ var_cov @ x_pred.T)

    ls_means.append({
        "Treatment": trt,
        "LS_Mean": ls_mean,
        "SE": se_pred,
        "CI_Lower": ls_mean - 1.96 * se_pred,
        "CI_Upper": ls_mean + 1.96 * se_pred
    })

ls_means_df = pl.DataFrame(ls_means)
# LS Means (95% CI)
ls_means_df
```

| Treatment<br>str       | LS_Mean<br>f64 | SE<br>f64 | CI_Lower<br>f64 | CI_Upper<br>f64 |
|------------------------|----------------|-----------|-----------------|-----------------|
| "Placebo"              | 0.071554       | 0.171563  | -0.264709       | 0.407818        |
| "Xanomeline Low Dose"  | -0.105215      | 0.171308  | -0.440977       | 0.230548        |
| "Xanomeline High Dose" | 0.388498       | 0.177055  | 0.041471        | 0.735525        |

## 10.8 Step 6: Pairwise treatment comparisons

We calculate treatment differences and their statistical significance.

```
# Calculate pairwise comparisons vs. placebo
tbl2_data = []
comparisons = [
    ("Xanomeline Low Dose vs. Placebo", "TRTP[T.Xanomeline Low Dose]"),
    ("Xanomeline High Dose vs. Placebo", "TRTP[T.Xanomeline High Dose]")
]

for comp_name, trt_coef in comparisons:
    # Extract coefficient estimates
    coef = model.params[trt_coef]
    se = model.bse[trt_coef]
    t_stat = coef / se
    df = model.df_resid
    p_value = 2 * (1 - scipy_stats.t.cdf(abs(t_stat), df))

    # Calculate confidence interval
    ci_lower = coef - scipy_stats.t.ppf(0.975, df) * se
    ci_upper = coef + scipy_stats.t.ppf(0.975, df) * se

    tbl2_data.append({
        "Comparison": comp_name,
        "Estimate": coef,
        "SE": se,
        "CI_Lower": ci_lower,
        "CI_Upper": ci_upper,
        "t_stat": t_stat,
        "p_value": p_value
    })
```

```

    })

comparison_df = pl.DataFrame(tbl2_data)
# Treatment comparisons vs. placebo
comparison_df

```

| Comparison<br>str                  | Estimate<br>f64 | SE<br>f64 | CI_Lower<br>f64 | CI_Upper<br>f64 | t_stat<br>f64 | p_value<br>f64 |
|------------------------------------|-----------------|-----------|-----------------|-----------------|---------------|----------------|
| "Xanomeline Low Dose vs. Placebo"  | -0.176769       | 0.242635  | -0.654862       | 0.301324        | -0.728539     | 0.467031       |
| "Xanomeline High Dose vs. Placebo" | 0.316943        | 0.246806  | -0.169369       | 0.803256        | 1.284179      | 0.200383       |

## 10.9 Step 7: Prepare tables for RTF output

We format the analysis results into publication-ready tables.

```

# Table 1: Descriptive Statistics and LS Means
tbl1_data = []
for s, ls in zip(desc_stats, ls_means):
    tbl1_data.append([
        s["Treatment"],
        str(s["N_Baseline"]),
        f"{s['Baseline_Mean']:.1f} ({s['Baseline_SD']:.2f})" if not np.isnan(s['Baseline_Mean']) else "",
        str(s["N_Week24"]),
        f"{s['Week24_Mean']:.1f} ({s['Week24_SD']:.2f})" if not np.isnan(s['Week24_Mean']) else "",
        str(s["N_Change"]),
        f"{s['Change_Mean']:.1f} ({s['Change_SD']:.2f})" if not np.isnan(s['Change_Mean']) else "",
        f"{ls['LS_Mean']:.2f} ({ls['CI_Lower']:.2f}, {ls['CI_Upper']:.2f})"
    ])

tbl1 = pl.DataFrame(tbl1_data, orient="row", schema=[
    "Treatment", "N_Base", "Mean_SD_Base", "N_Wk24", "Mean_SD_Wk24",
    "N_Chg", "Mean_SD_Chg", "LS_Mean_CI"
])

# Table 1 - Descriptive Statistics and LS Means
tbl1

```

| Treatment<br>str       | N_Base<br>str | Mean_SD_Base<br>str | N_Wk24<br>str | Mean_SD_Wk24<br>str | N_Chg<br>str | Mean_SD_C<br>str |
|------------------------|---------------|---------------------|---------------|---------------------|--------------|------------------|
| "Placebo"              | "79"          | "5.7 (2.23)"        | "79"          | "5.6 (1.65)"        | "79"         | "-0.0 (2.32)"    |
| "Xanomeline Low Dose"  | "79"          | "5.4 (0.95)"        | "79"          | "5.4 (1.06)"        | "79"         | "-0.1 (1.02)"    |
| "Xanomeline High Dose" | "74"          | "5.4 (1.37)"        | "74"          | "5.8 (2.21)"        | "74"         | "0.4 (1.65)"     |

```
# Table 2: Pairwise Comparisons (formatted for output)
tbl2_formatted = []
for row in tbl2_data:
    tbl2_formatted.append([
        row["Comparison"],
        f"{row['Estimate']:.2f} ({row['CI_Lower']:.2f}, {row['CI_Upper']:.2f})",
        f"{row['p_value']:.4f}" if row['p_value'] >= 0.0001 else "<0.0001"
    ])

tbl2 = pl.DataFrame(tbl2_formatted, orient="row", schema=["Comparison", "Diff_CI", "P_Value"])

# Table 2 - Pairwise Comparisons
tbl2
```

| Comparison<br>str                  | Diff_CI<br>str        | P_Value<br>str |
|------------------------------------|-----------------------|----------------|
| "Xanomeline Low Dose vs. Placeb... | "-0.18 (-0.65, 0.30)" | "0.4670"       |
| "Xanomeline High Dose vs. Place... | "0.32 (-0.17, 0.80)"  | "0.2004"       |

## 10.10 Step 8: Create regulatory-compliant RTF document

We generate a comprehensive efficacy table following regulatory submission standards.

```
# Create comprehensive RTF document with multiple table sections
doc_ancova = rtf.RTFDocument(
    df=[tbl1, tbl2],
    rtf_title=rtf.RTFTitle(
        text=[
```

```

        "ANCOVA of Change from Baseline in",
        "Fasting Glucose (mmol/L) at Week 24 (LOCF)",
        "Efficacy Analysis Population"
    ]
),
rtf_column_header=[
    # Header for descriptive statistics table
    [
        rtf.RTFColumnHeader(
            text=["", "Baseline", "Week 24 (LOCF)", "Change from Baseline", ""],
            col_rel_width=[3, 2, 2, 2, 2],
            text_justification=["l", "c", "c", "c", "c"],
            text_format="b"
        ),
        rtf.RTFColumnHeader(
            text=[
                "Treatment Group",
                "N", "Mean (SD)",
                "N", "Mean (SD)",
                "N", "Mean (SD)",
                "LS Mean (95% CI){^a}"
            ],
            col_rel_width=[3, 0.7, 1.3, 0.7, 1.3, 0.7, 1.3, 2],
            text_justification=["l"] + ["c"] * 7,
            border_bottom="single",
            text_format="b"
        )
    ],
    # Header for pairwise comparisons table
    [
        rtf.RTFColumnHeader(
            text=[
                "Pairwise Comparison",
                "Difference in LS Mean (95% CI){^a}",
                "p-Value{^b}"
            ],
            col_rel_width=[5, 4, 2],
            text_justification=["l", "c", "c"],
            text_format="b",
            border_bottom="single"
        )
    ]
]

```

```

    )
]
],
rtf_body=[
    # Body for descriptive statistics
    rtf.RTFBody(
        col_rel_width=[3, 0.7, 1.3, 0.7, 1.3, 0.7, 1.3, 2],
        text_justification=["l"] + ["c"] * 7
    ),
    # Body for pairwise comparisons
    rtf.RTFBody(
        col_rel_width=[5, 4, 2],
        text_justification=["l", "c", "c"]
    )
],
rtf_footnote=rtf.RTFFootnote(
    text=[
        "{^a}LS means and differences in LS means are based on an ANCOVA model with treatment effects",
        f"{{^b}}p-values are from the ANCOVA model testing treatment effects",
        "LOCF (Last Observation Carried Forward) approach is used for missing Week 24 values",
        "ANCOVA = Analysis of Covariance; CI = Confidence Interval; LS = Least Squares; SD = Standard Deviation"
    ]
),
rtf_source=rtf.RTFSource(
    text=[
        "Source: ADLBC Analysis Dataset",
        f"Analysis conducted: {pd.Timestamp.now().strftime('%d%b%Y').upper()}",
        "Statistical software: Python (statsmodels)"
    ]
)
)

# Generate RTF file
doc_ancova.write_rtf("rtf/tlf_efficacy_ancova.rtf")

```

rtf/tlf\_efficacy\_ancova.rtf

PosixPath('pdf/tlf\_efficacy\_ancova.pdf')

## **Part III**

# **Analysis package**

# 11 Packaging overview

## Objective

Understand the concept of analysis packages for clinical study reports. Learn how Python packages provide structure, reproducibility, and compliance for regulatory submissions.

## 11.1 What is an analysis package

An **analysis package** is a Python package designed specifically to organize analysis scripts and code for a clinical trial project.

Unlike general-purpose Python packages distributed on PyPI, analysis packages serve as:

- Project containers for clinical trial deliverables
- Reproducible environments for analyses
- Submission-ready structures for regulatory review

Think of it as combining:

- Python package structure (for code organization)
- Quarto project (for report generation)
- Regulatory requirements (for eCTD submission)

## 11.2 Why use an analysis package

Clinical trial projects have unique needs that standard Python projects may not address:

**Regulatory compliance:**



- FDA requires submission of analysis programs in ASCII text format
- Reviewers must be able to reproduce your results
- Documentation must explain the analysis process

**Team collaboration:**

- Multiple statisticians and programmers work on hundreds of tables
- Consistent structure reduces communication overhead
- Shared functions avoid code duplication

**Long-term maintenance:**

- Analysis must be reproducible years later
- Environment must be reconstructable
- Code and data provenance must be clear

The Python package structure addresses these needs systematically.

## 11.3 Analysis package vs standard package

Python packages serve different purposes depending on context.

**Standard Python package** (for PyPI):

- Purpose: Share reusable functionality
- Audience: General Python community
- Scope: Generic, broadly applicable functions
- Example: `polars`, `plotnine`, `rtflite`

**Analysis package** (for submissions):

- Purpose: Organize trial-specific analyses
- Audience: Study team and regulators
- Scope: Study-specific tables, listings, figures
- Example: `demo001` (DEMO-001 study analysis)

In R terms, think of an analysis package like a project-specific R package (e.g., `esubdemo`) versus a CRAN package (e.g., `dplyr`).

## 11.4 Key components

A typical analysis package contains:

### Python package structure:

- `pyproject.toml`: Project metadata and dependencies
- `src/studyname/`: Study-specific Python functions
- `tests/`: Validation and testing code
- `uv.lock`: Exact dependency versions

### Analysis content:

- `analysis/`: Quarto documents for TLFs
- `data/`: ADaM datasets (input)
- `output/`: Generated tables, listings, figures (output)

### Documentation:

- `README.md`: Project overview
- `_quarto.yml`: Quarto book configuration

This structure supports the complete lifecycle: development, validation, and submission.

## 11.5 Demo project

This book uses [demo-py-esub](https://github.com/elong0527/demo-py-esub) as the demonstration project.

The project shows how to:

- Organize analysis code as a Python package
- Generate clinical study reports with Quarto
- Prepare deliverables for eCTD submission

Clone the project to follow along:

```
git clone https://github.com/elong0527/demo-py-esub.git
cd demo-py-esub
```

The project generates six TLFs:

- Disposition of patients
- Study population
- Baseline characteristics
- Efficacy analysis (ANCOVA)
- Adverse events summary
- Adverse events (specific)

These cover the most common clinical reporting scenarios.

## 11.6 Workflow overview

The typical workflow for an analysis package:

### 1. Project setup:

- Initialize Python package with `uv`
- Configure Quarto for report generation
- Set up version control with Git

### 2. Development:

- Write analysis functions in `src/`
- Create Quarto documents in `analysis/`
- Generate TLFs in `output/`

### 3. Validation:

- Write tests in `tests/`
- Perform independent review
- Verify outputs match specifications

### 4. Submission:

- Pack package into text files with `pkglite`
- Place files in eCTD Module 5 structure
- Update ADRG with reproduction instructions

The following chapters detail each stage.

## 11.7 Benefits of this approach

Using Python packages for clinical analysis provides:

### Consistency:

- Standard structure across all projects
- Team members know where to find code and outputs
- Reduces onboarding time for new projects

### Automation:

- `uv` manages dependencies automatically
- Quarto renders all reports in batch
- Testing frameworks verify correctness

### Reproducibility:

- `uv.lock` ensures exact dependency versions
- `.python-version` specifies Python version
- Repository snapshots freeze package ecosystem

### Compliance:

- Built-in documentation with docstrings
- Testing infrastructure for validation
- Standard structure simplifies review

#### ! Important

For regulatory submissions, reproducibility is not optional. The FDA expects to reconstruct your exact environment and verify your results.

## 11.8 What's next

The next chapters cover:

- **Package structure:** Organizing code and content (Chapter [12](#))
- **Project management:** Git-centric workflows for collaboration (Chapter [13](#))

- **Submission overview:** eCTD requirements and pkglite (Chapter [14](#))
- **Submission package:** Packing for eCTD Module 5 (Chapter [15](#))
- **Submission dryrun:** Verifying reproducibility (Chapter [16](#))

With this foundation, you're ready to learn how to manage analysis packages effectively.

# 12 Package structure

## Objective

Learn the recommended directory structure for clinical analysis packages. Understand how to organize Python code, Quarto documents, data, and outputs in a reproducible, submission-ready layout.

## 12.1 Core principle

Organize clinical analysis projects as valid Python packages that leverage the entire Python packaging toolchain (especially `uv`).

Additionally, integrate essential components for clinical reporting:

- Quarto documents for reproducible analysis
- ADaM datasets as inputs
- Generated TLFs as outputs

This hybrid structure combines Python package best practices with clinical trial deliverable requirements.

## 12.2 Complete example structure

A clinical analysis package requires these essential components:

```
demo-py-esub/
  pyproject.toml
  .python-version
  uv.lock
  README.md
  .gitignore
  _quarto.yml
  index.qmd
  src/
    demo001/
      __init__.py
      utils.py
      baseline.py
      efficacy.py
      population.py
      safety.py
  analysis/
    tlf-01-disposition.qmd
    tlf-02-population.qmd
    tlf-03-baseline.qmd
    tlf-04-efficacy-ancova.qmd
    tlf-05-ae-summary.qmd
    tlf-06-specific.qmd
  data/
    adsl.parquet
    adae.parquet
    adlbc.parquet
    advs.parquet
    adtte.parquet
  output/
    tlf-disposition.rtf
    tlf-population.rtf
    tlf-baseline.rtf
    tlf-efficacy-ancova.rtf
    tlf-ae-summary.rtf
    tlf-ae-specific.rtf
  tests/
    __init__.py
    test_utils.py
    test_baseline.py
    data/
```

`adsl_subset.parquet`

This structure satisfies both Python packaging standards and regulatory submission requirements.

In R terms, this combines:

- R package structure  
(DESCRIPTION, R/, tests/)
- Analysis project layout  
(vignettes/, data/, output/)

## 12.3 Python package components

The Python package portion follows the [Python Packaging User Guide](#).

### 12.3.1 pyproject.toml

The single source of truth for project configuration:

```
[project]
name = "demo001"
version = "0.1.0"
description = "Analysis package for DEMO-001 study"
readme = "README.md"
requires-python = ">=3.13"
dependencies = [
    "plotnine>=0.15.1",
    "polars>=1.35.2",
    "rtflite>=1.1.0",
]

[dependency-groups]
dev = [
    "mypy>=1.18.2",
    "pytest>=9.0.1",
    "ruff>=0.14.5",
]

[build-system]
requires = ["uv_build>=0.12.5,<0.13.0"]
build-backend = "uv_build"
```

Key sections:



- `[project]`: Package metadata
- `[project].dependencies`: Runtime dependencies for analysis
- `[dependency-groups.dev]`: Development tools (testing, linting)
- `[build-system]`: How to build the package

### 12.3.2 `.python-version`

Specifies the exact Python version:

3.14.0

Created by `uv python pin 3.14.0`.

#### ! Important

Use the full MAJOR.MINOR.PATCH version (for example, 3.14.0), not just 3.14. This prevents drift as new patch versions are released.

### 12.3.3 `uv.lock`

Lock file with exact dependency versions:

```
version = 1
requires-python = ">=3.13"

[[package]]
name = "polars"
version = "1.35.1"
source = { registry = "https://pypi.org/simple" }
...
```

This file is auto-generated by `uv sync` and `uv lock`.

Never edit manually. Commit to version control.

### 12.3.4 src/demo001/

Study-specific Python functions go here.

Following the `src/` layout (recommended):

```
src/
  demo001/
    __init__.py      # Package initialization
    utils.py         # Utility functions
    baseline.py      # Baseline characteristics
    efficacy.py      # Efficacy analysis
    population.py    # Population analysis
    safety.py        # Safety analysis
```

#### Why `src/` layout?

- Prevents accidental imports from development directory
- Forces proper package installation
- Industry best practice

Each module contains related functions. For example, `utils.py`:

```
"""Utility functions for formatting and calculations."""

def fmt_num(x: float, digits: int = 1, width: int = 5) -> str:
    """Format a number with specified digits and width.

    Parameters
    -----
    x : float
        Number to format
    digits : int
        Number of decimal places
    width : int
        Total width of formatted string

    Returns
    -----
    str
```

Formatted number string

Examples

-----

```
>>> fmt_num(12.345, digits=2, width=6)
' 12.35'
"""
return f"{x:>{width}.{digits}f}"
```

Document all functions with docstrings following the [NumPy docstring standard](#).

### 12.3.5 tests/

Validation tests using pytest:

```
tests/
__init__.py
test_utils.py          # Test utility functions
test_baseline.py       # Test baseline functions
data/                  # Test data fixtures
    adsl_subset.parquet
```

Example test:

```
# tests/test_utils.py
from demo001.utils import fmt_num

def test_fmt_num_basic():
    """Test basic number formatting."""
    assert fmt_num(12.345, digits=2, width=6) == " 12.35"

def test_fmt_num_padding():
    """Test width padding."""
    assert fmt_num(1.2, digits=1, width=5) == " 1.2"
```

Run tests:

```
uv run pytest
```

### **i** Note

For clinical submissions, high test coverage demonstrates code quality. Aim for >80% coverage for critical functions.

## 12.4 Quarto project components

The Quarto portion enables reproducible report generation.

### 12.4.1 `_quarto.yml`

Quarto project configuration:

```
project:
  type: book

book:
  title: "DEMO-001 Analysis Results"
  chapters:
    - index.qmd
    - analysis/tlf-01-disposition.qmd
    - analysis/tlf-02-population.qmd
    - analysis/tlf-03-baseline.qmd
    - analysis/tlf-04-efficacy-ancova.qmd
    - analysis/tlf-05-ae-summary.qmd
    - analysis/tlf-06-specific.qmd

format:
  html:
    theme: cosmo
```

This configures Quarto to render all analysis documents as a book.

### 12.4.2 index.qmd

Landing page for the Quarto book:

```
---
title: "DEMO-001 Clinical Study Report"
---

## Overview

This analysis package contains Tables, Listings, and Figures (TLFs)
for the DEMO-001 clinical trial.

## Study Information

- Protocol: DEMO-001
- Phase: III
- Indication: [Disease]
- Primary Endpoint: [Endpoint]

## Analysis Programs

The following TLFs are included:

- **Disposition**: Patient disposition table
- **Population**: Analysis population summary
- **Baseline**: Baseline characteristics
- **Efficacy**: Primary efficacy analysis (ANCOVA)
- **AE Summary**: Adverse events summary
- **AE Specific**: Specific adverse events
```

### 12.4.3 analysis/

Analysis scripts as Quarto documents:

```
analysis/
  tlf-01-disposition.qmd
  tlf-02-population.qmd
  tlf-03-baseline.qmd
  tlf-04-efficacy-ancova.qmd
```

```
tlf-05-ae-summary.qmd
tlf-06-specific.qmd
```

Each .qmd file:

- Loads required data
- Performs analysis
- Generates formatted output
- Exports to RTF for submission

Example structure:

```
---
title: "Table 14.1.1 - Disposition of Patients"
---

## Load Data

```{python}
import polars as pl
from demo001.utils import fmt_num

adsl = pl.read_parquet("data/adsl.parquet")
```

## Analysis

```{python}
# Calculate disposition counts
disposition = adsl.group_by("TRTA").agg([
    pl.len().alias("N"),
    pl.col("EOSSTT").filter(pl.col("EOSSTT") == "COMPLETED").count().alias("Completed")
])
```

## Output

```{python}
from rtflite import Table
# Generate RTF table...
```
```

 Warning

For final submissions, convert .qmd files to .py scripts.  
Covered in Chapter [15](#).

## 12.5 Data and output directories

### 12.5.1 data/

Input datasets in Parquet format:

```
data/
  adsl.parquet      # Subject-level analysis dataset
  adae.parquet      # Adverse events
  adlbc.parquet     # Lab chemistry
  advs.parquet      # Vital signs
  adtte.parquet     # Time-to-event
```

#### Why Parquet?

- Faster than CSV for large datasets
- Preserves data types
- Smaller file size
- Python ecosystem standard

For submission, ADaM datasets are converted to SAS .xpt format per FDA requirements.

### 12.5.2 output/

Generated TLF outputs:

```
output/
  tlf-disposition.rtf
  tlf-population.rtf
  tlf-baseline.rtf
  tlf-efficacy-ancova.rtf
  tlf-ae-summary.rtf
  tlf-ae-specific.rtf
```

RTF files are submission-ready and can be converted to PDF for review.

## 12.6 Additional files

### 12.6.1 .gitignore

Exclude generated files from version control:

```
# Python
__pycache__/*
*.py[cod]
.venv/

# Quarto
_book/
*.html

# Output (generated)
output/

# OS
.DS_Store
Thumbs.db
```

The `output/` directory typically goes in `.gitignore` since outputs are generated from source code. Commit source, not generated artifacts.

### 12.6.2 README.md

Project documentation:

```
# demo-py-esub

Analysis package for DEMO-001 clinical trial.

## Installation

```bash
git clone https://github.com/org/demo-py-esub.git
```



```
cd demo-py-esub
uv sync
```

## Usage

Generate all TLFs:

```bash
quarto render
```

Run tests:

```bash
uv run pytest tests/
```
```

## 12.7 Benefits of this structure

### Consistency:

- Every project follows the same layout
- Team members instantly know where files belong
- Reduces cognitive load

### Reproducibility:

- `uv.lock` pins all dependencies
- `.python-version` specifies Python version
- Quarto renders from source every time

### Automation:

- `uv sync` restores environment
- `quarto render` regenerates all outputs
- `pytest` validates all functions

### Compliance:

- Standard Python package can be built and distributed

- Tests provide validation evidence
- Documentation is built-in

### Note

The structure scales well. A project with 300 TLFs uses the same layout, just more files in **analysis/** and **src/**.

## 12.8 Mixed language projects

If your organization uses both R and Python:

### Separate projects:

```
clinical-trial-001/
  r-package/           # R-based analyses
    DESCRIPTION
    R/
    vignettes/
  python-package/      # Python-based analyses
    pyproject.toml
    src/
    analysis/
  data/                # Shared ADaM datasets
  output/              # Shared outputs
```

### Why separate?

- Different build systems (devtools vs uv)
- Different dependency management (renv vs uv)
- Different testing frameworks (testthat vs pytest)
- Simpler to maintain

Share data and outputs, not source code.

 Mixing programming languages in a single project is often a mistake

John Carmack [noted](#): “It’s almost always a mistake to mix languages in a single project.”

Mixing languages in a single project increases the complexity of dependency management, testing, and build processes. It can lead to confusion about which tools to use for specific tasks and complicate collaboration among team members who may specialize in different languages. Keep Python projects pure Python. Keep R projects pure R. Share data and outputs, not codebases.

## 12.9 What's next

You've learned the recommended structure for analysis packages.

The next part covers eCTD submission:

- Regulatory requirements for program submission
- Using pkglite to pack Python packages
- Creating submission packages
- Verifying reproducibility with dry runs

With this structure in place, you're ready to prepare submission-ready deliverables.

# 13 Project management

## Objective

Learn Git-centric workflows for managing clinical analysis projects. Understand agile development practices and validation strategies for regulatory compliance.

## 13.1 Git-centric workflow

Clinical analysis projects require rigorous tracking and collaboration. A Git-centric workflow provides the foundation for reproducible, auditable work.

### **Core principle:**

All project assets live in version control. Work is tracked through issues, pull requests, and project boards.

This applies whether you use:

- GitHub Enterprise
- GitLab Self-Managed/Dedicated
- Bitbucket Data Center
- Azure DevOps

The specific platform matters less than the workflow discipline.

## 13.2 Plain text workflow

Favor plain text formats for all project artifacts:

**Use:**

- `.qmd` files for analysis scripts (not Jupyter notebooks for final deliverables)
- `.md` files for documentation
- `.toml` files for configuration
- `.txt` files for submission packages

**Avoid:**

- `.xlsx` files for tracking (no audit trail, merge conflicts)
- Binary formats when text alternatives exist
- Proprietary formats that require special tools

**i Note**

Plain text enables:

- Clear diff views in pull requests
- Meaningful merge conflict resolution
- Searchable code history
- Command-line friendly workflows

## 13.3 Project tracking

Use properly labeled issues and pull requests to drive work. Follow [Tidyteam code review principles](#) for detailed operational advices.

### 13.3.1 Issues for requirements

Create issues to capture:

- Individual TLF specifications
- Bug reports
- Feature requests
- Validation tasks

Example issue template:

```
**Title**: Implement Table 14.1.1 - Disposition of Patients

**Description**:
Create disposition table following ICH E3 guidelines

**Deliverables**:

- [ ] Quarto document: analysis/tlf-01-disposition.qmd
- [ ] Output table: output/tlf-disposition.rtf
- [ ] Unit tests: tests/test_disposition.py

**Validation**: Independent review required
```

### 13.3.2 Pull requests for review

**ALL** code changes must go through pull requests before getting into `main`:

- Developer creates feature branch
- Implements changes
- Opens pull request for review
- Reviewer validates code and outputs
- Changes merged after approval

This creates an audit trail of who did what and when.

### 13.3.3 Project boards

Use Kanban-style project boards to visualize work:

**Columns:**

- Backlog: Planned work
- In Progress: Active development
- Review: Awaiting validation
- Done: Completed and validated

**Example board:**

| Backlog       | In Progress  | Review       | Done         |
|---------------|--------------|--------------|--------------|
| Table 14.3.5  | Table 14.1.1 | Table 14.2.1 | Table 14.1.2 |
| Figure 14.4.1 |              | Table 14.3.1 | Table 14.2.2 |

This makes project status transparent to all stakeholders.

### 13.4 Development lifecycle

Clinical analysis projects follow a structured development life-cycle similar to software development.

#### 13.4.1 Planning

Define scope and requirements:

- List all TLFs from Statistical Analysis Plan (SAP)
- Create mock tables/shells
- Assign validation levels (independent review vs double programming)
- Set up validation tracking

Create a validation tracker (plain text format):

```
| TLF | Type | Developer | Dev Status | Reviewer | Review Status |
|-----|-----|-----|-----|-----|-----|
| tlf-01-disposition | Table | Alice | Complete | Bob | In Progress |
| tlf-02-population | Table | Charlie | In Progress | Diana | Pending |
```

**! Important**

Lock down the Python version and package repository snapshot during planning. Changing these mid-project breaks reproducibility.

GitHub Projects, GitLab Boards, and Jira all support this workflow. Choose based on your organization's infrastructure.

### 13.4.2 Development

Team members implement assigned TLFs:

#### 1. Create feature branch:

```
git checkout -b feature/tlf-01-disposition
```

#### 2. Implement analysis:

- Write Quarto document in `analysis/`
- Create helper functions in `src/` if needed
- Generate output in `output/`

#### 3. Self-test:

- Verify against mock table
- Check calculations manually
- Run automated tests

#### 4. Commit and push:

```
git add analysis/tlf-01-disposition.qmd  
git commit -m "Implement disposition table (Table 14.1.1)"  
git push origin feature/tlf-01-disposition
```

#### 5. Open pull request:

Request review from assigned validator.

### 13.4.3 Validation

Independent reviewers verify deliverables:

#### For tables:

- Compare output against specifications
- Verify calculations independently
- Check formatting requirements
- Review code for errors

#### For analysis functions:



- Write unit tests in `tests/`
- Verify edge cases
- Check type annotations
- Review docstrings

#### Validation testing example:

```
# tests/test_disposition.py
import polars as pl
from demo001.disposition import create_disposition_table

def test_disposition_counts():
    """Verify disposition table calculations."""
    # Load test data
    df = pl.read_parquet("tests/data/adsl_subset.parquet")

    # Generate table
    result = create_disposition_table(df)

    # Verify counts
    assert result["Screened"][0] == 254
    assert result["Randomized"][0] == 254
    assert result["Completed"][0] == 238
```

Run tests with pytest:

```
uv run pytest
```

### 13.4.4 Delivery

Project lead ensures completion:

#### 1. Run compliance checks:

```
uv run ruff check .
uv run mypy src/
uv run pytest --cov=demo001
```

#### 2. Generate all outputs in batch:

```
quarto render
```

### 3. Review validation tracker:

Ensure all TLFs have completed validation.

### 4. Prepare submission package:

Pack for eCTD (covered in Chapter 15).

## 13.5 Agile practices

Clinical trials benefit from agile project management:

### Iterative development:

- Work in 2-week sprints
- Deliver subset of TLFs each sprint
- Get stakeholder feedback early

### Continuous integration:

- Automated testing on every push
- Catch errors before review
- Maintain code quality

### Regular retrospectives:

- What went well?
- What needs improvement?
- Adjust process accordingly

#### **i** Note

Agile doesn't mean uncontrolled. The validation requirements remain the same. Agile provides faster feedback loops within those constraints.

## 13.6 Automation with CI/CD

Automate repetitive tasks using continuous integration:

**GitHub Actions example:**

```
name: Validation

on: [push, pull_request]

jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v5
      - uses: astral-sh/setup-uv@v7
      - run: uv sync
      - run: uv run pytest tests/
      - run: uv run ruff check .
      - run: uv run mypy src/
```

This runs tests automatically on every pull request.

**Benefits:**

- Catch errors before manual review
- Ensure consistent code quality
- Reduce reviewer burden
- Document test results

## 13.7 Collaboration best practices

**Work as a team:**

- Take project management training
- Understand your role in the lifecycle
- Communicate blockers early

**Design clean architecture:**

CI/CD is optional but highly recommended for projects with multiple developers.

- Separate business logic from data processing
- Write reusable components in `src/`
- Keep analysis scripts in `analysis/` focused

#### Set capability boundaries:

- Know what your team can deliver
- Avoid complex integrations (e.g., mixing R and Python in same package)
- Prefer simple, robust solutions

#### Contribute to community:

- Share reusable components internally
- Open source when possible
- Learn from others' approaches

## 13.8 Version control discipline

#### Branch strategy:

- `main`: Stable, validated code
- `feature/*`: Development branches
- `hotfix/*`: Emergency fixes

#### Commit messages:

Write clear, descriptive commits:

```
# Good
git commit -m "Add ANCOVA analysis for primary endpoint (Table 14.2.1)"

# Bad
git commit -m "Update code"
```

#### Never commit:

- Sensitive data
- Large binary files
- Generated outputs (use `.gitignore`)
- Temporary files

**Always commit:**

- Source code (`.py`, `.qmd`)
- Configuration (`pyproject.toml`, `_quarto.yml`)
- Documentation (`README.md`)
- Tests (`tests/*.py`)

## 13.9 What's next

You've learned Git-centric project management workflows.

The next chapter covers the detailed package structure:

- Directory layout for analysis packages
- Organizing code and content
- Integrating Quarto with Python packages
- Essential configuration files

These practices ensure consistent, reproducible, collaborative clinical analysis projects.

## **Part IV**

# **eCTD submission**

## 14 Submission overview

### Objective

Understand FDA requirements for submitting analysis programs. Learn about the eCTD structure and how pkglite enables Python package submissions.

### 14.1 Electronic Common Technical Document

The electronic Common Technical Document (eCTD) provides a standard format for regulatory submissions.

The eCTD organizes submission documents in a defined directory structure:

- **Module 1:** Administrative information and prescribing information
- **Module 2:** Common Technical Document summaries
- **Module 3:** Quality (CMC)
- **Module 4:** Nonclinical study reports
- **Module 5:** Clinical study reports

For analysis programs, we focus on **Module 5** (clinical study reports).

Full eCTD specifications are available from the [ICH website](#).

### 14.2 FDA requirements for analysis programs

The [FDA Study Data Technical Conformance Guide](#) (Section 4.1.2.10) specifies:

Sponsors should provide the software programs used to create all ADaM datasets and **generate tables and figures associated with primary and secondary efficacy analyses**. Furthermore, sponsors should submit software programs used to generate additional information included in Section 14 CLINICAL STUDIES of the Prescribing Information (PI) if applicable. **The specific software utilized should be specified in the ADRG**. The main purpose of requesting the submission of these programs **is to understand the process by which the variables for the respective analyses were created and to confirm the analysis algorithms**. Sponsors should submit software programs in **ASCII text format**; however, executable file extensions should not be used.

Key requirements:

1. Submit programs for primary and secondary efficacy analyses
2. Specify software and versions in ADRG
3. Use ASCII text format
4. No executable extensions

## 14.3 eCTD Module 5 structure

Analysis datasets and programs are organized under Module 5:

```
m5/datasets/<study-id>/analysis/adam/
```

Within the `adam/` folder, two directories are critical:

```
m5/datasets/<study-id>/analysis/adam/
  datasets/
    *.xpt           # ADaM datasets in SAS format
    define.xml      # Dataset definitions
    adrg.pdf        # Analysis Data Reviewer's Guide
```



```
analysis-results-metadata.pdf # Analysis Results Metadata
programs/
  py0pkgs.txt                # Packed Python packages
  tlf-01-disposition.txt      # Analysis program 1
  tlf-02-population.txt       # Analysis program 2
  ...                        # Additional programs
```

## 14.4 The ASCII text requirement

Why ASCII text?

**Platform independence:**

- Works on any operating system
- No special software needed to view
- Future-proof format

**Review process:**

- Reviewers can read code without running it
- Easy to search and navigate
- Can copy code snippets for testing

**Compliance verification:**

- Plain text prevents hidden code
- No macros or embedded executables
- Transparent to automated scanning

This creates a challenge: how to submit a Python package (which has directory structure, binary files, etc.) as ASCII text files?

## 14.5 The solution: **pkglite** for Python

[pkglite for Python](#) solves the text file requirement by packing Python projects into portable text files. Key capabilities:

- Pack entire project directory structure into single text file
- Preserve file paths and metadata
- Exclude unnecessary files with `.pkgliteignore`

- Unpack to restore original structure
- Support multiple projects in one file

**i** Note

pkglite for Python extends the original pkglite for R with:

- Support for any programming language (not just R)
- Content-based file classification (text vs binary)
- Command-line interface for automation
- `.pkgliteignore` configuration support

### 14.5.1 How pkglite works

**Packing:**

1. Scan project directory
2. Classify files (text vs binary)
3. Encode file paths and contents
4. Write to single `.txt` file

**Unpacking:**

1. Read `.txt` file
2. Parse file paths and contents
3. Recreate directory structure
4. Write files to disk

The packed text file follows the Debian Control File (DCF) format, similar to the R package pkglite output.

## 14.6 Python language considerations

As of the [August 30, 2025 FDA guidance update](#), the eCTD Module 5 specification explicitly allows `.zip` files “for delivering R packages.”

However, Python (and other languages) are not explicitly mentioned.

**Our approach:**

Use pkglite to pack Python packages into portable text files.  
This follows the **spirit** of the FDA guidance:

- Provides ASCII text format programs
- Enables reproducibility
- Documents software versions
- Allows reviewer verification

## 14.7 Submission workflow overview

The complete submission workflow:

### 1. Develop analysis package:

- Create Python package with uv
- Write analysis code in Quarto documents
- Validate outputs and functions

### 2. Prepare submission package:

- Pack Python package with pkglite
- Convert Quarto documents to Python scripts
- Place files in eCTD Module 5 structure

### 3. Update documentation:

- Update ADRG with software versions
- Provide reproduction instructions
- Update ARM with program metadata

### 4. Verify reproducibility:

- Perform dry run test
- Unpack and install packages
- Reproduce analysis results

The next chapters detail steps 2-4.

We developed pkglite for Python specifically to enable submission of source projects in **any** programming language following the same principles.

## 14.8 What goes in the submission

**Python packages** (`programs/py0pkgs.txt`):

All study-specific Python packages. These contain helper functions used across multiple analyses.

**Analysis programs** (`programs/tlf-*.txt`):

Individual analysis scripts. Each generates one or more TLFs.

**ADaM datasets** (`datasets/*.xpt`):

Analysis datasets in SAS transport format. Required by CDISC standards.

**Documentation** (`datasets/adrg.pdf`, `datasets/analysis-results-metadata.pdf`):

ADRG provides:

- Python version and package versions
- Reproduction instructions
- Platform requirements

ARM provides:

- Links between programs and outputs
- Program metadata
- Analysis descriptions

## 14.9 Dependencies and package management

**Public packages:**

Packages available on PyPI (e.g., `polars`, `rtflite`, `pkglite`) do not need submission. Document versions in ADRG.

**Proprietary packages:**

Internal packages (e.g., company-specific utilities) should be included if they are:

- Hosted in private repositories (not public)
- Required to run the analyses

- Not available to reviewers

Pack proprietary packages together with analysis packages using `pkglite`.

#### **Python version:**

Specify the exact Python version in ADRG. Use `uv python pin` to lock the version in the project.

#### **Package snapshots:**

If your organization uses a package snapshot (similar to Posit Package Manager), provide the snapshot date in ADRG.

## **14.10 Platform considerations**

#### **Cross-platform compatibility:**

Python code should work on Windows, macOS, and Linux.

Avoid:

- Platform-specific paths (`C:\` vs `/usr/`)
- Platform-specific system calls
- Binary dependencies that require compilation

Use:

- `pathlib` for path handling
- Pure Python packages when possible
- Wheels for platform-independent distribution

#### **External dependencies:**

Minimize external dependencies beyond Python packages.

If required (e.g., system libraries), document clearly in ADRG.

#### **Warning**

Each external dependency increases the complexity of environment recreation. Keep it simple.

## 14.11 Next steps

The following chapters provide detailed instructions for:

- **Submission package** (Chapter [15](#)): Using pkglite to pack Python packages and analysis programs. Converting Quarto documents to Python scripts. Organizing files in eCTD structure.
- **Submission dryrun** (Chapter [16](#)): Simulating the reviewer experience. Unpacking and installing packages. Reproducing analysis results.

With this foundation, you're ready to prepare your first Python submission package.

# 15 Submission package

## Objective

Learn how to pack Python analysis packages into eCTD-ready text files. Know the workflow for converting Quarto documents to Python scripts and organizing files in Module 5 structure.

## 15.1 Prerequisites

This chapter uses two demo repositories:

**Analysis project:** [demo-py-esub](https://github.com/elong0527/demo-py-esub)

```
git clone https://github.com/elong0527/demo-py-esub.git
```

**Submission package:** [demo-py-ectd](https://github.com/elong0527/demo-py-ectd)

```
git clone https://github.com/elong0527/demo-py-ectd.git
```

We assume paths `demo-py-esub/` and `demo-py-ectd/` below.

### Install pkglite:

pkglite is not needed as a project dependency. Use `uvx` to run without installation:

```
uvx pkglite --help
```

Or install globally with `pipx`:

```
pipx install pkglite
```

### **i** Note

`uvx` and `pipx` both run command-line tools in isolated environments. Use `uvx` for one-off commands or `pipx` for frequently used tools.

## 15.2 The whole game

The eCTD Module 5 structure for Python submissions:

```
demo-py-ectd/m5/datasets/ectddemo/analysis/adam/  
  datasets/  
    adsl.xpt  
    adae.xpt  
    ...  
    adrg.pdf  
    analysis-results-metadata.pdf  
    define.xml  
    define2-0-0.xsl  
  programs/  
    py0pkgs.txt  
    tlf-01-disposition.txt  
    tlf-02-population.txt  
    tlf-03-baseline.txt  
    tlf-04-efficacy.txt  
    tlf-05-ae-summary.txt  
    tlf-06-ae-specific.txt
```

**datasets/ folder:**

- ADaM datasets in `.xpt` format (SAS transport)
- `define.xml` metadata (created by Pinnacle 21 or similar tools)
- ADRG and ARM documentation

**programs/ folder:**



- `py0pkgs.txt`: Packed Python package(s)
- `tlf-*.txt`: Individual analysis programs

All files in `programs/` must be ASCII text files.

#### ! Important

File naming conventions:

- Use lowercase letters only
- No underscores or special characters
- Use hyphens for separators
- No executable extensions (`.py`, `.exe`)

## 15.3 Packing Python packages with py-pkglite

### 15.3.1 Create `.pkgliteignore`

First, create a `.pkgliteignore` file to exclude unnecessary files:

```
uvx pkglite use demo-py-esub/
```

Created `.pkgliteignore` in `/home/user/demo-py-esub`

This generates `demo-py-esub/.pkgliteignore` with default exclusions:

```
# Git
.git/

# OS
.DS_Store
Thumbs.db

# Python
__pycache__/
```

```

*.py[cod]
.venv/

# R
.Rproj.user/
.Rhistory
.RData
.Ruserdata

# Quarto
.quarto/

```

The `.pkgliteignore` file follows gitignore syntax.

#### What to exclude:

- Generated files (`__pycache__`, `.venv`)
- Data and outputs (submitted separately)
- Version control metadata
- Quarto book content (if converted to scripts separately)

#### What to include:

- Source code (`src/`)
- Configuration (`pyproject.toml`, `uv.lock`, `.python-version`)
- Tests (`tests/`)
- Essential documentation for package installation

### 15.3.2 Pack the package

Pack the analysis package into a text file:

```

uvx pkglite pack demo-py-esub/ \
  -o demo-py-ectd/m5/datasets/ectddemo/analysis/adam/programs/py0pkgs.txt

```

```

Packing demo-py-esub
Reading _quarto.yml
Reading uv.lock

```

pkglite uses content-based file classification. Text files are packed directly. Binary files trigger warnings.

```
Reading .pkgliteignore
Reading pyproject.toml
Reading index.qmd
Reading README.md
Reading .gitignore
Reading .python-version
Reading analysis/tlf-05-ae-summary.qmd
Reading analysis/tlf-02-population.qmd
Reading analysis/tlf-03-baseline.qmd
Reading analysis/.gitignore
Reading analysis/tlf-06-specific.qmd
Reading analysis/tlf-01-disposition.qmd
Reading analysis/tlf-04-efficacy-ancova.qmd
Reading tests/test_utils.py
Reading tests/__init__.py
Reading output/tlf_ae_specific.rtf
Reading output/tlf_baseline.rtf
Reading output/tlf_population.rtf
Reading output/tlf_disposition.rtf
Reading output/tlf_ae_summary.rtf
Reading output/tlf_efficacy_ancova.rtf
Reading .github/.gitignore
Reading .github/workflows/quarto-publish.yml
Reading data/adae.parquet
Reading data/adlbhy.parquet
Reading data/adsl.parquet
Reading data/adtte.parquet
Reading data/adlbc.parquet
Reading data/adlbh.parquet
Reading data/advs.parquet
Reading src/demo001/baseline.py
Reading src/demo001/population.py
Reading src/demo001/__init__.py
Reading src/demo001/utils.py
Reading src/demo001/safety.py
Reading src/demo001/efficacy.py
Packed 1 packages into /home/user/demo-py-ectd/m5/datasets/ectddemo/analysis/adam/programs/py
```

This creates a single text file containing:

- pyproject.toml

- `.python-version`
- `uv.lock`
- All files in `src/`
- All files in `tests/`

### 15.3.3 Inspect the packed file

View the first few lines:

```
head -n 20 demo-py-ectd/m5/datasets/ectddemo/analysis/adam/programs/py0pkgs.txt
```

```
# Generated by py-pkg-lite: do not edit by hand
# Use `pkg-lite unpack` to restore the packages
```

```
Package: demo-py-esub
```

```
File: _quarto.yml
```

```
Format: text
```

```
Content:
```

```
  project:
    type: book
```

```
  book:
```

```
    title: "DEMO-001 Analysis Results"
```

```
    chapters:
```

- `index.qmd`
- `analysis/tlf-01-disposition.qmd`
- `analysis/tlf-02-population.qmd`
- `analysis/tlf-03-baseline.qmd`
- `analysis/tlf-04-efficacy-ancova.qmd`
- `analysis/tlf-05-ae-summary.qmd`
- `analysis/tlf-06-specific.qmd`

### 15.3.4 Packing multiple packages

If you have dependencies in private repositories:

```
uvx pkg-lite pack internal-utils/ demo-py-esub/ \
-o demo-py-ectd/m5/datasets/ectddemo/analysis/adam/programs/py0pkgs.txt
```

Packages are packed in order specified. Always pack dependencies first (low-level before high-level).

**i** Note

Unpacking will restore packages in the same order. Depending on how you reinstall them, the order may matter.

## 15.4 Converting Quarto to Python scripts

Analysis programs must be plain Python scripts, not Quarto documents.

### 15.4.1 The conversion workflow

For each `.qmd` file in `analysis/`:

1. Render to verify it works
2. Convert `.qmd` to `.ipynb` (Jupyter notebook)
3. Convert `.ipynb` to `.py` (Python script)
4. Clean up comments and formatting
5. Save as `.txt` file

### 15.4.2 Automated conversion script

This shell script automates the conversion:

```
#!/bin/bash

cd demo-py-esub/
uv sync
source .venv/bin/activate

convert_analysis() {
    local analysis_name=$1
    local analysis_path="analysis/$analysis_name.qmd"
    local output_path="$HOME/demo-py-ectd/m5/datasets/ectddemo/analysis/adam/programs"
```

```

# Render .qmd to verify it works
quarto render "$analysis_path" --quiet

# Convert .qmd to .ipynb
quarto convert "$analysis_path"

# Convert .ipynb to .py using nbconvert
uvx --from nbconvert jupyter-nbconvert \
    --to python "analysis/$analysis_name.ipynb" \
    --output "$output_path/$analysis_name.py"

# Remove all comments (lines starting with #)
awk '!/^#/' "$output_path/$analysis_name.py" > temp && \
    mv temp "$output_path/$analysis_name.py"

# Consolidate consecutive blank lines
awk 'NF {p = 0} !NF {p++} p < 2' "$output_path/$analysis_name.py" > temp && \
    mv temp "$output_path/$analysis_name.py"

# Clean up intermediate files
rm "analysis/$analysis_name.ipynb"

# Format with ruff
uvx ruff format "$output_path/$analysis_name.py"

# Rename .py to .txt (no executable extension)
mv "$output_path/$analysis_name.py" "$output_path/$analysis_name.txt"
}

# Convert all analysis files
for qmd_file in analysis/*.qmd; do
    analysis=$(basename "$qmd_file" .qmd)
    convert_analysis "$analysis"
done

```

Save as `convert_analyses.sh` and run:

```

chmod +x convert_analyses.sh
./convert_analyses.sh

```

### 15.4.3 Add reviewer instructions

Optionally, add a header to each .txt file:

```
# Note to Reviewer
#
# To rerun this analysis program, please refer to the ADRG appendix.
```

This helps reviewers understand how to execute the code.

Automated insertion:

```
header='# Note to Reviewer
#
# To rerun this analysis program, please refer to the ADRG appendix.
#
'

add_header() {
    local file=$1
    echo "$header" | cat - "$file" > temp && mv temp "$file"
}

for txt_file in demo-py-ectd/m5/datasets/ectddemo/analysis/adam/programs/tlf-*.txt; do
    add_header "$txt_file"
done
```

## 15.5 Verifying ASCII compliance

Non-ASCII characters (curly quotes, em dashes, Unicode symbols) will cause submission issues. Always verify before submission.

Currently, there is no built-in py-pkg-lite utility to check for ASCII compliance, so contributions are welcome! An example verification script is available in rtfliite: [verify\\_ascii.py](#).

## 15.6 Compliance checklist

Before finalizing the submission package:

File naming:

- [ ] All filenames are lowercase
- [ ] No underscores or special characters
- [ ] No `.py` extensions (use `.txt`)

Content:

- [ ] All `.txt` files are ASCII compliant
- [ ] Python package unpacks correctly
- [ ] Analysis programs run without errors

Structure:

- [ ] Files in correct eCTD Module 5 directories
- [ ] `pyOpkgs.txt` in `programs/`
- [ ] Analysis programs in `programs/`
- [ ] ADaM datasets in `datasets/`

Documentation:

- [ ] ADRG includes Python version
- [ ] ADRG includes package versions
- [ ] ADRG includes reproduction instructions
- [ ] ARM links programs to outputs

## 15.7 Updating ADRG

The ADRG must document the Python environment and provide reproduction instructions.

### 15.7.1 Section: Macro Programs

Example content:



## 7.X Macro Programs

Submitted Python programs follow the naming pattern `tlf-##-\*.txt`.  
All study-specific Python functions are saved in the `py0pkgs.txt` file.

The recommended steps to unpack and use these functions are described in the Appendix.

The table below contains the software version and program metadata:

**\*\*Analysis programs table:\*\***

| Program Name           | Output Table | Title                      |
|------------------------|--------------|----------------------------|
| tlf-01-disposition.txt | Table 14.1.1 | Disposition of Patients    |
| tlf-02-population.txt  | Table 14.1.2 | Analysis Population        |
| tlf-03-baseline.txt    | Table 14.1.3 | Baseline Characteristics   |
| tlf-04-efficacy.txt    | Table 14.2.1 | Efficacy Analysis (ANCOVA) |
| tlf-05-ae-summary.txt  | Table 14.3.1 | Adverse Events Summary     |
| tlf-06-ae-specific.txt | Table 14.3.2 | Specific Adverse Events    |

**\*\*Python environment table:\*\***

| Software | Version | Description          |
|----------|---------|----------------------|
| Python   | 3.14.0  | Programming language |
| uv       | 0.9.9   | Package manager      |

**\*\*Python packages table:\*\***

| Package     | Version | Description          |
|-------------|---------|----------------------|
| polars      | 1.35.1  | Data manipulation    |
| plotnine    | 0.15.1  | Data visualization   |
| rtflite     | 1.1.0   | RTF table generation |
| statsmodels | 0.14.0  | Statistical models   |

**\*\*Proprietary packages table:\*\***

| Package | Version | Description |
|---------|---------|-------------|
|---------|---------|-------------|

```
| demo001 | 0.1.0 | DEMO-001 study analysis functions |
```

## 15.7.2 Appendix: Reproduction instructions

Package versions can be extracted from `uv.lock` or by running `uv pip list`.

Provide step-by-step instructions for reviewers:

### Appendix: Instructions to Execute Analysis Programs

#### 1. Install uv

Follow instructions at <https://docs.astral.sh/uv/getting-started/installation/>

#### 2. Create working directory

Create a temporary directory (e.g., ``C:/tempwork/`` on Windows).

Copy all files from ``m5/datasets/ectddemo/analysis/adam/`` to this directory.

#### 3. Unpack and install Python packages

Navigate to the working directory and run:

```
```bash
uvx pkglite unpack programs/py0pkgs.txt -o .
```
```

This restores the package structure in the current directory.

Install the package:

```
```bash
cd demo-py-esub
uv sync
```
```

This installs all dependencies and the `demo001` package.

#### 4. Copy data to the correct location

Ensure the `datasets/` folder with ADaM datasets is in the working directory.

## 5. Execute analysis programs

Run each program in order:

```
```bash
cd demo-py-esub
source .venv/bin/activate      # macOS/Linux
# or .venv\Scripts\activate   # Windows

python ../programs/tlf-01-disposition.txt
python ../programs/tlf-02-population.txt
python ../programs/tlf-03-baseline.txt
python ../programs/tlf-04-efficacy.txt
python ../programs/tlf-05-ae-summary.txt
python ../programs/tlf-06-ae-specific.txt
```
```

Each program generates RTF output in the specified output directory.

### Note

Tailor the instructions to your organization's environment. Include any special configuration (e.g., proxy settings, internal package indexes).

## 15.8 Updating ARM

The Analysis Results Metadata (ARM) documents the relationship between programs and outputs.

### 15.8.1 Section 2: Analysis Results Metadata Summary

Example table:

| Table Reference | Table Title              | Programming Language | Program Name           | Input Files |
|-----------------|--------------------------|----------------------|------------------------|-------------|
| Table 14.1.1    | Disposition of Patients  | Python               | tlf-01-disposition.txt | adsl.xpt    |
| Table 14.1.2    | Analysis Population      | Python               | tlf-02-population.txt  | adsl.xpt    |
| Table 14.1.3    | Baseline Characteristics | Python               | tlf-03-baseline.txt    | adsl.xpt    |

## 15.8.2 Section 3: Analysis Results Metadata Details

For each table, provide:

```
Table Reference: Table 14.1.1
Analysis Result: Disposition counts by treatment group
Analysis Reason: Describe study population
Analysis Purpose: Primary
Programming Statements: (Python version 3.14.0), [programs/tlf-01-disposition.txt]
```

## 15.9 Testing the submission package

Before finalizing, test the complete workflow:

1. Unpack `py0pkgs.txt` in a clean directory
2. Install packages per ADRG instructions
3. Run each analysis program
4. Verify outputs match original results

This is covered in detail in Chapter [16](#).

## 15.10 What's next

You've learned how to prepare Python submission packages.

The next chapter covers dry run testing:

- Simulating the reviewer workflow
- Unpacking and installing from text files
- Reproducing analysis results
- Verifying compliance

Dry run testing ensures your submission package works correctly.

# 16 Submission dryrun

## Objective

Learn how to perform a dry run test of your submission package. Simulate the reviewer experience to verify that your analysis can be reproduced from the submitted materials.

## 16.1 Why dry run testing

Before submitting to regulatory agencies, you must verify that reviewers can reproduce your analysis.

A dry run test simulates the reviewer workflow:

1. Start with a clean environment
2. Follow ADRG instructions exactly
3. Unpack and install packages
4. Run analysis programs
5. Verify outputs match original results

This catches issues before submission:

- Missing dependencies
- Incorrect file paths
- Platform-specific code
- Documentation gaps
- Version mismatches

## Important

Dry run testing is not optional. It's the only way to ensure your submission package actually works.

## 16.2 Prerequisites

You need the eCTD submission package prepared in Chapter 15:

```
git clone https://github.com/elong0527/demo-py-ectd.git
cd demo-py-ectd
```

The structure:

```
demo-py-ectd/m5/datasets/ectddemo/analysis/adam/
  datasets/
    *.xpt
    adrg.pdf
    analysis-results-metadata.pdf
  programs/
    py0pkgs.txt
    tlf-*.txt
```

## 16.3 Setting up the test environment

### 16.3.1 Create a clean directory

Simulate a reviewer's fresh environment:

```
# Create temporary directory
mkdir ~/dryrun-test
cd ~/dryrun-test

# Copy submission materials
cp -r ~/demo-py-ectd/m5/datasets/ectddemo/analysis/adam/* .

# Verify structure
ls -R
```

You should see:

datasets programs

./datasets:

|            |             |             |           |            |
|------------|-------------|-------------|-----------|------------|
| adadas.xpt | adlbc.xpt   | adlbhvp.xpt | adrg.pdf  | advs.xpt   |
| adae.xpt   | adlbcpv.xpt | adlbhy.xpt  | adsl.xpt  | define.xml |
| adcibc.xpt | adlbh.xpt   | adnpix.xpt  | adtte.xpt | define2-   |
| 0-0.xsl    |             |             |           |            |

./programs:

|                        |                            |      |
|------------------------|----------------------------|------|
| py0pkgs.txt            | tlf-03-baseline.txt        | tlf- |
| 06-specific.txt        |                            |      |
| tlf-01-disposition.txt | tlf-04-efficacy-ancova.txt |      |
| tlf-02-population.txt  | tlf-05-ae-summary.txt      |      |

#### Note

Use a completely separate directory, not your development environment. This ensures you are testing from a clean state.

### 16.3.2 Install uv

Follow the ADRG instructions exactly.

```
# Verify
uv --version
```

```
uv 0.9.7 (0adb44480 2025-10-30)
```

## 16.4 Unpacking the Python package

### 16.4.1 Unpack with py-pkglite

Use pkglite to restore the package structure:

```
uvx pkglite unpack programs/py0pkgs.txt -o .
```



```
Unpacking demo-py-esub
Writing _quarto.yml
Writing uv.lock
Writing .pkgliteignore
Writing pyproject.toml
Writing index.qmd
Writing README.md
Writing .gitignore
Writing .python-version
Writing analysis/tlf-05-ae-summary.qmd
Writing analysis/tlf-02-population.qmd
Writing analysis/tlf-03-baseline.qmd
Writing analysis/.gitignore
Writing analysis/tlf-06-specific.qmd
Writing analysis/tlf-01-disposition.qmd
Writing analysis/tlf-04-efficacy-ancova.qmd
Writing tests/test_utils.py
Writing tests/__init__.py
Writing output/tlf_ae_specific.rtf
Writing output/tlf_baseline.rtf
Writing output/tlf_population.rtf
Writing output/tlf_disposition.rtf
Writing output/tlf_ae_summary.rtf
Writing output/tlf_efficacy_ancova.rtf
Writing .github/.gitignore
Writing .github/workflows/quarto-publish.yml
Writing data/adae.parquet
Writing data/adlbhy.parquet
Writing data/adsl.parquet
Writing data/adtte.parquet
Writing data/adlbc.parquet
Writing data/adlbh.parquet
Writing data/advs.parquet
Writing src/demo001/baseline.py
Writing src/demo001/population.py
Writing src/demo001/__init__.py
Writing src/demo001/utils.py
Writing src/demo001/safety.py
Writing src/demo001/efficacy.py
  Unpacked 1 packages from programs/py0pkgs.txt into .
```

This should create:

```
demo-py-esub
  _quarto.yml
  analysis
    tlf-01-disposition.qmd
    tlf-02-population.qmd
    tlf-03-baseline.qmd
    tlf-04-efficacy-ancova.qmd
    tlf-05-ae-summary.qmd
    tlf-06-specific.qmd
  data
    adae.parquet
    adlbc.parquet
    adlbh.parquet
  ...
```

Verify the structure:

```
ls -la demo-py-esub/
```

Or:

```
tree demo-py-esub/
```

The unpacked directory name comes from the packed package. It should match the original analysis project name.

## 16.5 Installing dependencies

### 16.5.1 Sync environment

Navigate to the unpacked package and install:

```
cd demo-py-esub
uv sync
```

This creates:

- `.venv/` virtual environment
- Installs all dependencies from `uv.lock`
- Installs the `demo001` package in editable mode

## 16.5.2 Verify installation

Check Python version in virtual environment:

```
source .venv/bin/activate # macOS/Linux
# .venv\Scripts\activate # Windows

python --version
```

List installed packages:

```
uv pip list
```

Verify the demo001 package:

```
python -c "import demo001; print(demo001.__version__)"
```

The output should be 0.1.0.

### Note

If any imports fail, check:

- Python version matches `.python-version`
- All dependencies installed from `uv.lock`
- Package installed in editable mode

## 16.6 Running analysis programs

### 16.6.1 Execute programs

Run each analysis program:

```
# Run first program
python ../programs/tlf-01-disposition.txt
```

You should see:

```
/home/user/dryrun-test/demo-py-esub/output/tlf_disposition.rtf
```

The program should:

1. Load data from ../datasets/
2. Perform analysis
3. Generate RTF output in output/

Run remaining programs:

```
python ../programs/tlf-02-population.txt
python ../programs/tlf-03-baseline.txt
python ../programs/tlf-04-efficacy-ancova.txt
python ../programs/tlf-05-ae-summary.txt
python ../programs/tlf-06-specific.txt
```

## 16.6.2 Verify outputs

Check that all RTF files were created:

```
ls -lh output/
```

```
total 456
-rw-r--r--@ 1 user  staff   170K Nov  6 21:59 tlf_ae_specific.rtf
-rw-r--r--@ 1 user  staff    9.6K Nov  6 21:59 tlf_ae_summary.rtf
-rw-r--r--@ 1 user  staff    7.6K Nov  6 21:58 tlf_baseline.rtf
-rw-r--r--@ 1 user  staff    14K Nov  6 21:57 tlf_disposition.rtf
-rw-r--r--@ 1 user  staff    8.7K Nov  6 21:59 tlf_efficacy_ancova.rtf
-rw-r--r--@ 1 user  staff    4.0K Nov  6 21:58 tlf_population.rtf
```

## 16.7 Comparing outputs

### 16.7.1 Manual comparison

Open RTF files in a word processor and compare with original outputs, check if:

- Table structure matches

- Numbers are identical
- Formatting is correct
- Headers and footers present

Since RTF files are plaintext files, you can also automate this comparison with diff-based workflows.

## 16.8 Testing checklist

Complete dry run test checklist:

Environment setup:

- [ ] Clean directory created
- [ ] Correct Python version installed
- [ ] uv installed and working

Package restoration:

- [ ] `pyOpkgs.txt` unpacked successfully
- [ ] Package structure restored
- [ ] All files present

Dependency installation:

- [ ] `uv sync` completed without errors
- [ ] All packages installed
- [ ] Correct package versions

Program execution:

- [ ] All analysis programs run without errors
- [ ] RTF outputs generated
- [ ] No missing data errors

Output verification:

- [ ] All expected outputs present
- [ ] Numbers match original results
- [ ] Tables formatted correctly

```
- [ ] No corruption or errors in RTF files
```

Documentation:

```
- [ ] ADRG instructions followed successfully  
- [ ] No undocumented steps required  
- [ ] Instructions are clear and complete
```

For projects with many programs, you can automate the dry run with a shell script. We will leave that as an exercise for the reader.

## 16.9 Best practices

### **Test early, test often:**

Don't wait until the last minute. Run dry run tests throughout development.

### **Test on fresh systems:**

Use virtual machines or Docker containers for truly clean environments.

### **Document everything:**

Record commands, outputs, and any issues encountered.

### **Version control dry run scripts:**

Keep test scripts in version control alongside analysis code.

### **Automate where possible:**

Automated tests are faster and more reliable than manual testing.

### **Test on reviewer platforms:**

If you know reviewers use Windows, test on Windows.

## 16.10 What's next

Congratulations! You've learned the complete workflow for Python-based clinical submissions. You're now ready to apply these practices to real clinical trial projects. The Python clinical reporting ecosystem continues to evolve. Stay engaged with the community to learn best practices and contribute improvements. Here we list the important resources to explore:

### Caution

In case you used GitHub Codespaces, remember to [stop \(and maybe delete\) your Codespace](#) to avoid unnecessary usage.

### Regulatory guidance:

- [FDA Study Data Technical Conformance Guide](#)
- [eCTD specifications](#)

### Technical documentation:

- [rtflite documentation](#)
- [pkglite for Python documentation](#)
- [uv documentation](#)
- [Python Packaging User Guide](#)

### Example repositories:

- [demo-py-esub](#): Analysis package
- [demo-py-ectd](#): Submission package

## References

Wickham, Hadley, and Jennifer Bryan. 2023. *R Packages*.  
O'Reilly Media, Inc.